Pair of Straight Lines (1)

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General Equation of the second degree:

The equation of the form, when a, b and h are not simultaneously zero,

$$ax^2+2hxy+by^2+2gx+2fy+c=0 \quad \text{---------} (1)$$

is called the general equation of the second degree or the quadratic equation in x and y.

Homogeneous Equation:

An equation of the form $f(x,y)=0$ is said to be the homogeneous equation of degree n, where n is a positive integer, and if for some real number k, we have

$$f(kx,ky)=k^n f(x,y)$$

For example, the equation $f(x,y)=x^4-3x^3y+9x^2y^2+8y^4$ is a homogeneous equation of degree 4, since

$$\begin{aligned} f(kx,ky) &= (kx)^4-3(kx)^3ky+9(kx)^2(ky)^2+8(ky)^4 \\ &= k^4x^4-3k^4x^3y+9k^4x^2y^2+8k^4y^4 \\ &= k^4\left[x^4-3x^3y+9x^2y^2+8y^4\right] \\ &= k^4 f(x,y) \end{aligned}$$

But the general second degree equation of the form:

$f(x,y)=ax^2+2hxy+by^2+2gx+2fy+c=0$ is not

homogeneous equation, since

$$f(kx,ky) = a(kx)^2 + 2h(kx)(ky) + b(ky)^2 + 2g(kx) + 2f(ky) + c$$

$$= ak^2x^2 + 2hk^2xy + bk^2y + 2kgx + 2kfy + c$$

$$= k^2\left[ax^2 + 2hxy + by^2 + 2\frac{g}{k}x + 2\frac{f}{k}y + \frac{c}{k^2}\right]$$

$$\neq k^2 f(x,y)$$

Second degree homogeneous equation:

An equation of the form $ax^2 + 2hxy + by^2 = 0$ is called the second degree homogeneous equation.

Since, $f(x,y) = ax^2 + 2hxy + by^2$

$$\therefore f(kx,ky) = a(kx)^2 + 2h(kx)(ky) + b(ky)^2$$

$$= k^2ax^2 + k^2(2hxy) + k^2by^2$$

$$= k^2\left[ax^2 + 2hxy + by^2\right]$$

$$= k^2 f(x,y)$$

Pair of Straight Lines

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Theorem:

Prove that a homogeneous equation of the second degree always represents a pair of straight lines through the origin.

Proof:

Let the homogeneous equation of second degree be

$$ax^2 + 2hxy + by^2 = 0 \quad \text{——— } ①$$

Multiplying both sides of ① by a (if $a \neq 0$) we obtain

$$a^2x^2 + 2haxy + aby^2 = 0$$

$$\Rightarrow (ax)^2 + 2 \cdot ax \cdot hy + (hy)^2 - (hy)^2 + aby^2 = 0$$

$$\Rightarrow (ax + hy)^2 - (h^2 - ab)y^2 = 0$$

$$\Rightarrow (ax + by)^2 - \left\{\left(\sqrt{h^2 - ab}\right)y\right\}^2 = 0$$

$$\Rightarrow \left\{ax + by + \left(\sqrt{h^2 - ab}\right)y\right\}\left\{ax + by - \left(\sqrt{h^2 - ab}\right)y\right\} = 0$$

$\therefore$ Either, $ax + by + \left(\sqrt{h^2 - ab}\right)y = 0$

or, $ax + by - \left(\sqrt{h^2 - ab}\right)y = 0$ ——— ②

which represents two straight lines, real or imaginary through the origin.

Remark:

These two straight lines represented by eqn (2) a

i) Real and different if $h^2 > ab$

ii) Real and coincident if $h^2 = ab$

iii) Imaginary if $h^2 < ab$.

Problem:

Find the angle between the two straight lines represented by the homogeneous equation of second degre

$ax^2 + 2hxy + by^2 = 0$.

Solution:

Given,

$$ax^2 + 2hxy + by^2 = 0 \quad \text{——— (1)}$$

$$\Rightarrow y^2 + 2\frac{h}{b}xy + \frac{a}{b}x^2 = 0 \quad \text{——— (2)}$$

Let $y - m_1 x = 0$ & $y - m_2 x = 0$ be the straight lines represented by eqn (1)

Now, from (2) we can write

$$y^2 + 2\frac{h}{b}xy + \frac{a}{b}x^2 \equiv (y - m_1 x)(y - m_2 x)$$

$$\Rightarrow y^2 + \frac{2h}{b}xy + \frac{a}{b}x^2 \equiv y^2 - (m_1 + m_2)xy + m_1 m_2 x^2$$

——— (3)

Equating the like terms from both sides of ③ we obtain

$$\left.\begin{aligned} m_1 + m_2 &= -\frac{2h}{b} \\ m_1 m_2 &= a/b \end{aligned}\right\} \quad \text{——— ④}$$

where m_1 & m_2 are the slopes of the straight lines.

Let, φ be the angle between the two lines, then

$$\tan\varphi = \frac{m_1 - m_2}{1 + m_1 m_2} \quad \text{——— ⑤}$$

Now,

$$(m_1 - m_2)^2 = (m_1 + m_2)^2 - 4m_1 m_2$$

$$= \left(-\frac{2h}{b}\right)^2 - \frac{4a}{b} \quad [\text{using ④}]$$

$$= \frac{4h^2}{b^2} - \frac{4a}{b}$$

$$= \frac{4h^2 - 4ab}{b^2}$$

$$= 4\left(\frac{h^2 - ab}{b^2}\right)$$

$$\therefore m_1 - m_2 = \frac{2\sqrt{h^2 - ab}}{b} \quad \text{——— ⑥}$$

$$⑤ \Rightarrow \tan\varphi = \frac{2\sqrt{h^2 - ab}}{b} * \frac{1}{1 + a/b}$$

$$\Rightarrow \tan\varphi = \frac{2\sqrt{h^2-ab}}{b} \times \frac{b}{a+b}$$

$$\therefore \tan\varphi = \frac{2\sqrt{h^2-ab}}{a+b} \quad \text{---------- (4)}$$

Corollary:

(1) If h^2-ab is positive i.e., $h^2-ab>0$, the lines are real, being coincident if $h^2-ab=0$

(2) If h^2-ab is negative, the lines are imaginary, but pass through the real point $(0,0)$.

(3) If $a+b=0$, i.e., if the sum of the coefficient of x^2 and y^2 is zero, $\tan\varphi$ is undefined, hence $\varphi = \pi/2$. Thus, the two lines are at right angles.

Problem:

Find the condition that the general equation of second degree $ax^2+2hxy+by^2+2gx+2fy+c=0$ may represent a pair of straight lines.

Solution:

Let the general equ^n of second degree be of the form

$$ax^2+2hxy+by^2+2gx+2fy+c=0 \quad \text{——— (1)}$$

represent a pair of straight lines and their point of intersection be (α, β).

Transfer the origin to (α,β) keeping the direction of axes same. Let (x', y') be the coordinates of any point on the locus (1) referred to the new axes.

Then

$$x = x'+\alpha \ \& \ y = y'+\beta \quad \text{——— (2)}$$

From eqn (1) we have using (2)

$$a(x'+\alpha)^2+2h(x'+\alpha)(y'+\beta)+b(y'+\beta)^2+2g(x'+\alpha)$$
$$+2f(y'+\beta)+c=0$$

$$\Rightarrow a(x'^2+2\alpha x'+\alpha^2)+2h(x'y'+\alpha y'+\beta x'+\alpha\beta)+b(y'^2+2\beta y'+\beta^2)$$
$$+2gx'+2g\alpha+2fy'+2f\beta+c=0$$

$$\Rightarrow ax'^2+2a\alpha x'+a\alpha^2+2hx'y'+2h\alpha y'+2h\beta x'+2h\alpha\beta$$
$$+by'^2+2b\beta y'+b\beta^2+2gx'+2g\alpha+2fy'+2f\beta+c=0$$

$$\Rightarrow ax'^2+2hx'y'+by'^2+2x'(a\alpha+h\beta+g)+2y'(h\alpha+b\beta+f)$$
$$+(a\alpha^2+2h\alpha\beta+b\beta^2+2g\alpha+2f\beta+c)=0 \quad \text{——— (3)}$$

Since, eqn (3) represent a pair of straight lines through

the new origin if must be homogeneous in x' and

Therefore, from (3) we get

$$a\alpha + h\beta + g = 0 \quad \text{——} (4)$$

$$h\alpha + b\beta + f = 0 \quad \text{——} (5)$$

and $a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0$ —— (6)

From (6) we can write

$$a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0$$

$$\Rightarrow \alpha(a\alpha + h\beta + g) + \beta(h\alpha + b\beta + f) + g\alpha + f\beta + c = 0$$

$$\Rightarrow \alpha \cdot 0 + \beta \cdot 0 + g\alpha + f\beta + c = 0 \quad [\text{using (4) \& (5)}]$$

$$\Rightarrow g\alpha + f\beta + c = 0 \quad \text{——} (7)$$

Eliminating α, β from (4), (5) & (7) we have

$$\Delta \equiv \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$$\Rightarrow a(bc - f^2) - h(ch - gf) + g(hf - bg) = 0$$

$$\Rightarrow abc - af^2 - ch^2 + fgh + fgh - bg^2 = 0$$

$$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

which is the required condition.

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Point of intersection of the two straight lines:

The point of intersection of the two straight lines represented by $ax^2+2hxy+by^2+2gx+2fy+c=0$

Let, (α, β) be the point of intersection, then (α,β) satisfy the equations (4) & (5)

$$\left.\begin{aligned} a\alpha+h\beta+g=0 \\ h\alpha+b\beta+f=0 \end{aligned}\right\} \longrightarrow (*)$$

Solving the system of equations (*) for α,β using cross multiplication, we have

$$\frac{\alpha}{hf-bg} = \frac{\beta}{gh-af} = \frac{1}{ab-h^2}$$

$$\therefore \alpha = \frac{hf-bg}{ab-h^2} \quad \& \quad \beta = \frac{gh-af}{ab-h^2}$$

If the lines are parallel then $ab-h^2=0$, therefore the point of intersection is at infinity.

Note:

Let, $F(x,y) = ax^2+2hxy+by^2+2gx+2fy+c$

$$\therefore F(\alpha,\beta) = a\alpha^2+2h\alpha\beta+b\beta^2+2g\alpha+2f\beta+c$$

$$\therefore \frac{\partial F}{\partial \alpha} = 2a\alpha+2h\beta+2g = 2(a\alpha+h\beta+g)$$

$$\therefore \frac{\partial F}{\partial \beta} = 2h\alpha + 2b\beta + 2f = 2(h\alpha + b\beta + f)$$

$$\left.\begin{aligned} \frac{\partial F}{\partial \alpha} = 0 &\Rightarrow a\alpha + h\beta + g = 0 \\ \frac{\partial F}{\partial \beta} = 0 &\Rightarrow h\alpha + b\beta + f = 0 \end{aligned}\right\}$$

So, the point of intersection is obtained by solving

$$\frac{\partial F}{\partial \alpha} = 0 \quad \& \quad \frac{\partial F}{\partial \beta} = 0$$

Theorem:

If the general equation of the second degree

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \text{——— (1)}$$

represents a pair of straight lines, then the equation

$$ax^2 + 2hxy + by^2 = 0 \quad \text{——— (2)}$$

represents a pair of straight lines parallel to them through the origin.

Proof:

Let $l_1x + m_1y + n_1 = 0$ ——— (3)

& $l_2x + m_2y + n_2 = 0$ ——— (4) be the straight lines represented by (1). Then we have

$$(l_1x + m_1y + n_1)(l_2x + m_2y + n_2) \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

$$\Rightarrow l_1l_2x^2 + m_1l_2xy + n_1l_2x + l_1m_2xy + m_1m_2y^2 + n_1m_2y$$
$$+ l_1n_2x + m_1n_2y + n_1n_2 \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

$$\Rightarrow l_1l_2x^2 + (m_1l_2 + l_1m_2)xy + m_1m_2y^2 + (n_1l_2 + n_2l_1)x$$
$$+ (m_1n_2 + m_2n_1)y + n_1n_2 \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c \quad \text{——— (5)}$$

Equating the coefficients of the like terms from both sides of (5) we obtain

$$l_1l_2 = a\,;\quad l_1m_2 + l_2m_1 = 2h\,;\quad m_1m_2 = b \quad \text{——— (6)}$$

Now, the straight lines through the origin and parallel to (3) & (4) are

$l_1x + m_1y = 0$ & $l_2x + m_2y = 0$, respectively

Therefore, their joint equation is

$$(l_1x + m_1y)(l_2x + m_2y) = 0$$

$$\Rightarrow l_1l_2x^2 + l_1m_2xy + l_2m_1xy + m_1m_2y^2 = 0$$

$$\Rightarrow l_1l_2x^2 + (l_1m_2 + l_2m_1)xy + m_1m_2y^2 = 0$$

$$\Rightarrow ax^2 + 2hxy + by^2 = 0 \quad [\text{using (6)}]$$

Hence, the theorem is proved.

Problem:

Find the angle between the straight lines represented by

$$ax^2+2hxy+by^2+2gx+2fy+c=0 \quad \text{——} (1)$$

Sol^n:

Let, $l_1x+m_1y+n_1=0$ —— (2)

& $l_2x+m_2y+n_2=0$ —— (3) are the straight lines represented by (1)

Therefore,

$$(l_1x+m_1y+n_1)(l_2x+m_2y+n_2) \equiv ax^2+2hxy+by^2+2gx+2fy+c$$

$$\Rightarrow l_1l_2x^2+(l_1m_2+l_2m_1)xy+m_1m_2y^2+(l_1n_2+l_2n_1)x$$

$$+(m_1n_2+m_2n_1)y+n_1n_2 \equiv ax^2+2hxy+by^2+2gx+2fy+c \quad \text{——} (4)$$

Equating the coefficients of the like terms from both sides of (4) we obtain

$$\left.\begin{aligned} l_1l_2=a;\quad l_1m_2+l_2m_1=2h;\quad m_1m_2=b \\ l_1n_2+l_2n_1=2g;\quad m_1n_2+m_2n_1=2f;\quad n_1n_1=c \end{aligned}\right\}$$

Now,

Slope of eq^n (2) is, $s_1=-l_1/m_1$

Slope of eq^n (3) is, $s_2=-l_2/m_2$

Let, ϕ be the angle between lines (2) & (3), then

$$\tan\phi = \frac{-S_1 - S_2}{1 + S_1 S_2} \qquad \left[\because \tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2}\right]$$

$$\Rightarrow \tan\phi = \frac{-l_1/m_1 - l_2/m_2}{1 + \frac{l_1 l_2}{m_1 m_2}} = \frac{\frac{m_2 l_1 - m_1 l_2}{m_1 m_2}}{\frac{m_1 m_2 + l_1 l_2}{m_1 m_2}}$$

$$\Rightarrow \tan\phi = \frac{l_1 m_2 - l_2 m_1}{l_1 l_2 + m_1 m_2}$$

$$\Rightarrow \tan\phi = \frac{\sqrt{(l_1 m_2 + l_2 m_1)^2 - 4 l_1 l_2 m_1 m_2}}{l_1 l_2 + m_1 m_2}$$

$$\Rightarrow \tan\phi = \frac{\sqrt{(2h)^2 - 4ab}}{a+b}$$

$$\Rightarrow \tan\phi = \frac{\sqrt{4h^2 - 4ab}}{a+b} = \frac{2\sqrt{h^2 - ab}}{a+b}$$

$$\therefore \tan\phi = \frac{2\sqrt{h^2 - ab}}{a+b} \qquad \text{——— (5)}$$

or,
$$\boxed{\phi = \tan^{-1}\left(\frac{2\sqrt{h^2 - ab}}{a+b}\right)}$$

which represents the angle between the straight lines represented by (2) & (3).

Remark:

- These two lines are real and distinct if $h^2 - ab > 0$
- These two lines are imaginary if $h^2 - ab < 0$
- These two lines are parallel if $h^2 - ab = 0$
- These two lines are perpendicular if $a + b = 0$

Since, the condition of two lines being parallel to each other is $h^2 - ab = 0$

$$\Rightarrow h \cdot h = a \cdot b$$

$$\boxed{\Rightarrow \frac{a}{h} = \frac{h}{b}} \quad \text{---------- (6)}$$

Again, the condition of the general equation of second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ representing a pair of straight lines be

$$\Delta = 0$$

$$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\Rightarrow \cancel{ch^2} + 2fg\sqrt{ab} - af^2 - bg^2 - \cancel{ch^2} = 0$$

$$\Rightarrow af^2 - 2\sqrt{ab}\, fg + bg^2 = 0$$

$$\Rightarrow \left(\sqrt{a}f - \sqrt{b}g\right)^2 = 0$$

$$\Rightarrow \sqrt{a}f - \sqrt{b}g = 0$$

$\Rightarrow \sqrt{a}f = \sqrt{b}g$

$\Rightarrow \sqrt{a}\cdot\sqrt{b}f = \sqrt{b}\cdot\sqrt{b}g$

$\Rightarrow \sqrt{ab}f = bg$

$\Rightarrow hf = bg \quad [\because h^2 = ab]$

$\Rightarrow \boxed{\frac{h}{b} = \frac{g}{f}}$ ——— (7)

Combining (6) & (7) the condition of parallelism of the lines (2) & (3) can be rewritten as

$$\boxed{\frac{a}{h} = \frac{h}{b} = \frac{g}{f}} \quad \text{——— (8)}$$

Problem:

Find the equation of the bisectors of the angle between the straight lines represented by

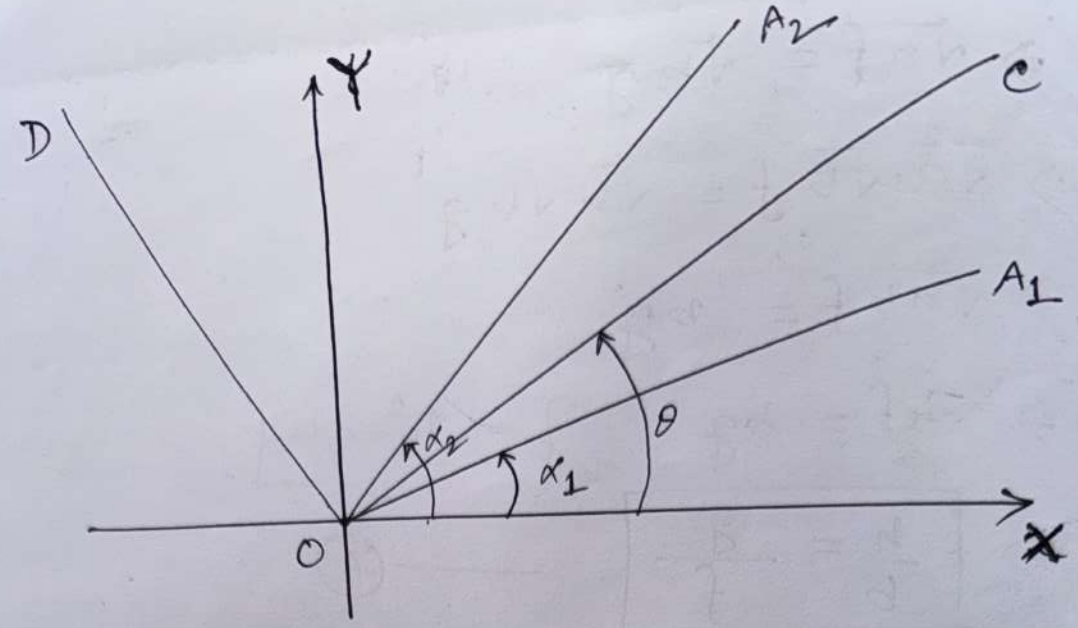
$$ax^2 + 2hxy + by^2 = 0 \quad \text{——— (1)}$$

Soln:

Let OA_1 and OA_2 be the straight lines represented by (1).
Let their separate equations be

$$y - m_1 x = 0 \quad \&$$

$$y - m_2 x = 0$$



Then we can write,

$$y^2 + \frac{2h}{b}xy + \frac{a}{b}x^2 \equiv (y - m_1x)(y - m_2x)$$

$$\Rightarrow y^2 + \frac{2h}{b}xy + \frac{a}{b}x^2 \equiv y^2 - (m_1 + m_2)x + m_1m_2x^2 \quad \text{——} (2)$$

which implies that

$$m_1 + m_2 = -\frac{2h}{b}$$
$$\& \quad m_1m_2 = a/b$$

Let these two lines make angles α_1 and α_2, respectively with X-axis. Therefore,

$$m_1 = \tan\alpha_1$$
$$m_2 = \tan\alpha_2 \quad \text{——} (3)$$

Let OC and OD be respectively the internal and external bisectors of the angle $\angle A_1OA_2$

Therefore, we can write

$$\angle A_1OC = \angle A_2OC \quad \text{——} (4)$$

Now,

$$\angle XOC = \angle XOA_1 + \angle A_1OC$$

$$= \angle XOA_1 + \frac{1}{2}\angle A_1OA_2 \quad [\text{using } ④]$$

$$= \angle XOA_1 + \frac{1}{2}(\angle XOA_2 - \angle XOA_1)$$

$$= \alpha_1 + \frac{1}{2}(\alpha_2 - \alpha_1)$$

$$\Rightarrow \angle XOC = \frac{1}{2}(\alpha_1 + \alpha_2)$$

$$\Rightarrow 2\angle XOC = \alpha_1 + \alpha_2 \quad \text{——— } ⑤$$

Again from elementary geometry, $\angle COD = \pi/2$

$$\therefore \angle XOD = \angle XOC + \angle COD = \frac{1}{2}(\alpha_1 + \alpha_2) + \pi/2$$

$$\Rightarrow 2\angle XOD = \pi + (\alpha_1 + \alpha_2) \quad \text{——— } ⑥$$

Hence, if θ be the angle which either bisector makes with the x-axis, i.e., if $\theta = \angle XOC$ or $\angle XOD$, then from ⑤ & ⑥ we can write

$$\tan 2\theta = \tan(\alpha_1 + \alpha_2) \quad [\because \tan(\pi + \alpha_1 + \alpha_2) = \tan(\alpha_1 + \alpha_2)]$$

$$\Rightarrow \frac{2\tan\theta}{1 - \tan^2\theta} = \frac{\tan\alpha_1 + \tan\alpha_2}{1 - \tan\alpha_1 \tan\alpha_2}$$

$$\Rightarrow \frac{2\tan\theta}{1 - \tan^2\theta} = \frac{m_1 + m_2}{1 - m_1 m_2} \quad [\text{using } ③]$$

$$\Rightarrow \frac{2\tan\theta}{1-\tan^2\theta} = \frac{-2h/b}{1-a/b} \quad [\text{using } ②]$$

$$\Rightarrow \frac{2\tan\theta}{1-\tan^2\theta} = \frac{2h}{a-b} \quad \text{——} ⑦$$

If (x, y) be any point on either bisector, then

$\tan\theta = y/x$ —— ⑧

using equⁿ ⑧, equⁿ ⑦ can be written as

$$\frac{2\,y/x}{1-y^2/x^2} = \frac{2h}{a-b}$$

$$\Rightarrow \frac{y/x}{\frac{x^2-y^2}{x^2}} = \frac{h}{a-b}$$

$$\Rightarrow \boxed{\frac{x^2-y^2}{xy} = \frac{a-b}{h}} \quad \text{——} ⑨$$

Thus, equⁿ ⑨ being a relation between the coordinates of any point on either side, is the joint equation of the two bisectors.

Corollary:

Let the pair of straight lines be

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \text{——} ⑩$$

intersecting at (α, β)

Now, shifting the origin to (α, β) and keeping the directions of the axes unchanged so that

$$\left.\begin{aligned} x &= x' + \alpha \\ y &= y' + \beta \end{aligned}\right\} \quad \text{—— (2)}$$

Thus, the equⁿ (1) reduces to the form

$$ax'^2 + 2hx'y' + by'^2 = 0 \quad \text{—— (3)}$$

whose bisectors of angles are given by

$$\frac{x'^2 - y'^2}{x' * y'} = \frac{a-b}{h}$$

Hence, going back to old axes through (2), the above equⁿ of bisectors of angles between the lines represented by (1) is given by

$$\boxed{\frac{(x-\alpha)^2 - (y-\beta)^2}{(x-\alpha)(y-\beta)} = \frac{a-b}{h}}$$

or, $$\boxed{\frac{(x-\alpha)^2 - (y-\beta)^2}{a-b} = \frac{(x-\alpha)(y-\beta)}{h}}$$

Problem:

Prove that the following equations represent a pair of straight lines.

(i) $x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$

(ii) $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$

(iii) $3x^2 + 10xy + 8y^2 + 16x + 26y + 21 = 0$

(iv) $2y^2 + 3xy - 5y - 6x + 2 = 0$

(v) $12x^2 + 7xy - 12y^2 - x + 7y - 1 = 0$

Sol^n:

(i) Given, $x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$ ——— (1)

Comparing eqn^n (1) with the general equation of second degree we obtain

$$a = 1;\ h = -5/2;\ b = -6,\ g = 7;\ f = 5/2;\ c = 4$$

Now,

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \Rightarrow \Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$\Rightarrow abc \Rightarrow \Delta = (1)(-6)(4) + 2(5/2)(7)(-5/2) - (1)(5/2)^2 - (-6)(7)^2 - (4)(-5/2)^2$

$$\Rightarrow \Delta = -24 - \frac{175}{2} - \frac{25}{4} + 294 - \frac{100}{4}$$

(ii) Given,

$$3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$$

$$\Rightarrow 3x^2 + 8xy - 3y^2 + 29x - 3y + 18 = 0 \quad \text{——— (1)}$$

Comparing equⁿ (1) with the general equⁿ of second degree we obtain,

$$a = 3;\ h = 4;\ b = -3;\ g = \frac{29}{2};\ f = -3/2;\ c = 18.$$

Now,

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

$$\Rightarrow \Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$\Rightarrow \Delta = (3)(-3)(18) + 2(-3/2)\left(\frac{29}{2}\right)(4) - 3(-3/2)^2 - (-3)\left(\frac{29}{2}\right)^2 - (18)(4)^2$$

$$\Rightarrow \Delta = -162 - 174 - \frac{27}{4} + \frac{2523}{4} - 288$$

$$= 624 - 624$$

$$\therefore \Delta = 0$$

Since, $\Delta = 0$, so the given equⁿ (1) represents a pair of straight lines.

(Proved)

(iii) Given,

$$3x^2 + 10xy + 8y^2 + 16x + 26y + 21 = 0 \quad \text{---------- (1)}$$

Comparing equⁿ (1) with the standard/general equⁿ of second degree we obtain

$$a = 3;\ h = 5;\ b = 8;\ g = 8;\ f = 13;\ c = 21$$

Now,

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

$$\Rightarrow \Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= (3)(8)(21) + 2(13)(8)(5) - 3(13)^2 - 8(8)^2 - 21(5)^2$$

$$= 504 + 1040 - 507 - 512 - 525$$

$$\therefore \Delta = 0$$

Since, $\Delta = 0$, the given equⁿ (1) represents a pair of straight lines.

(Proved)

Problem:

Find for what value of λ, the following equations represent a pair of straight lines.

(i) $12x^2+36xy+\lambda y^2+6x+6y+3=0$

(ii) $\lambda x^2+4xy+y^2-4x-2y-3=0$

(iii) $\lambda x^2-10xy+12y^2+5x-16y-3=0$

(iv) $2x^2-y^2+xy-2x-5y+\lambda=0$

(v) $\lambda xy-8x+9y-12=0$

Soln:

(i) Given, $12x^2+36xy+\lambda y^2+6x+6y+3=0$ ——— (1)

Comparing equ^n (1) with the general equ^n of second degree we obtain,

$a=12;\ h=18;\ b=\lambda;\ g=3;\ f=3;\ c=3$

Equ^n (1) will represent a pair of straight lines if

$$\Delta=\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}=0$$

$\Rightarrow abc+2fgh-af^2-bg^2-ch^2=0$

$\Rightarrow (12)(\lambda)(3)+2(3)(3)(18)-12(3)^2-\lambda(3)^2-3(18)^2=0$

$\Rightarrow 36\lambda+324-108-9\lambda-972=0$

$\Rightarrow 27\lambda = 756$

$\therefore \lambda = 28$

Ans

(ii) Given,

$$\lambda x^2 + 4xy + y^2 - 4x - 2y - 3 = 0 \quad \text{——— (1)}$$

Comparing equ^n (1) with the general equ^n of second degree we obtain,

$a = \lambda$; $h = 2$; $b = 1$; $g = -2$, $f = -1$, $c = -3$

Equ^n (1) will represent a pair of straight lines if

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$\Rightarrow \lambda(1)(-3) + 2(-1)(-2)(2) - \lambda(-1)^2 - 1(-2)^2 - (-3)(2)^2 = 0$

$\Rightarrow -3\lambda + 8 - \lambda - 4 + 12 = 0$

$\Rightarrow 4\lambda = 16$

$\therefore \lambda = 4$

Ans

(iii) Given,

$$\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0 \quad \text{——— (1)}$$

Comparing equⁿ ① with the general equⁿ of second degree, we obtain,

$$a = \lambda;\ h = -5;\ b = 12;\ g = 5/2,\ f = -8;\ c = -3$$

Now, equⁿ ① will represent a pair of straight lines if

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\Rightarrow \lambda(12)(-3) + 2(-8)(5/2)(-5) - \lambda(-8)^2 - 12(5/2)^2 - (-3)(-5)^2 = 0$$

$$\Rightarrow -36\lambda + 200 - 64\lambda - 75 + 75 = 0$$

$$\Rightarrow 100\lambda = 200$$

$$\Rightarrow \lambda = 2.$$

Ans

(iv) Given,

$$2x^2 - y^2 + xy - 2x - 5y + \lambda = 0 \quad \text{——— ①}$$

Comparing equⁿ ① with the general equⁿ of second degree we obtain

$$a = 2;\ h = \frac{1}{2};\ b = -1;\ g = -1,\ f = -5/2,\ c = \lambda$$

Now, equⁿ ① will represent a pair of straight lines if

a.

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$\Rightarrow (2)(-1)\lambda + 2(-5/2)^2 - (-1)(-1)^2 - \lambda(\frac{1}{2})^2 + 2(-5/2)(-1)(\frac{1}{2}) = 0$

$\Rightarrow -2\lambda + \frac{25}{2} + 1 - \frac{\lambda}{4} + \frac{5}{2} = 0$

$\Rightarrow \frac{9\lambda}{4} = \frac{27}{2} + \frac{5}{2} = 16 = -9$

$\Rightarrow \lambda = \frac{4}{9} * \frac{27}{2} = 6 \Rightarrow \lambda = -9 * (\frac{4}{9})$

$\therefore \lambda = -4$

(Ans)

(v) Given,

$$2xy - 8x + 9y - 12 = 0 \quad \text{---} \quad (1)$$

Comparing equⁿ (1) with the general equⁿ of second degree we obtain,

$a = 0;\ h = 2/2;\ b = 0,\ g = -4;\ f = 9/2;\ c = -12$

Now, equⁿ (1) will represent a pair of straight lines if

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$\Rightarrow 0 + 2(9/2)(-4)(\lambda/2) - 0 - 0 - (-12)(\lambda/2)^2 = 0$

$\Rightarrow -18\lambda + 3\lambda^2 = 0$

$\Rightarrow 3\lambda(\lambda - 6) = 0$

$\Rightarrow \lambda - 6 = 0 \quad [\because \lambda \neq 0]$

$\therefore \lambda = 6$

(Ans)

Problem:

Prove that the equation $8x^2 - 14xy + 6y^2 + 2x - y - 1 = 0$ represents a pair of straight lines. Also find their point of intersection, angle between the lines, and the equation of the bisectors of the angles between them.

Soln:

Given, $8x^2 - 14xy + 6y^2 + 2x - y - 1 = 0$ ——— (1)

Comparing equⁿ (1) with the general equation of second degree we have

$a = 8;\ h = -7;\ b = 6;\ g = 1;\ f = -\frac{1}{2};\ c = -1;$

Equⁿ (1) will represent a pair of straight lines if

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

Now,

$$abc + 2fgh - af^2 - bg^2 - ch^2 = (8)(6)(-1) + 2\left(-\tfrac{1}{2}\right)(1)(-7) - (8)\left(-\tfrac{1}{2}\right)^2 - 6(1)^2 - (-1)(-7)^2$$

$$= -48 + 7 - 2 - 6 + 49$$

$$= 0$$

Since, $\Delta = 0$, so eqn ① represents a pair of straight lines.

Let, (α, β) be their point of intersection, and also let $F(\alpha, \beta) = 8\alpha^2 - 14\alpha\beta + 6\beta^2 + 2\alpha - \beta - 1$

$$\therefore \frac{\partial F}{\partial \alpha} = 16\alpha - 14\beta + 2 = 2(8\alpha - 7\beta + 1) = 0 \quad \text{—— ②}$$

$$\& \quad \frac{\partial F}{\partial \beta} = -14\alpha + 12\beta - 1 = 0 \quad \text{—— ③}$$

Solving ② & ③ we obtain

$$\alpha = 5/2 \ \& \ \beta = 3.$$

Hence, the point of intersection is $(5/2, 3)$ Ans

Let, ϕ be the angle between the lines, then

$$\tan\phi = \frac{2\sqrt{h^2 - ab}}{a + b} = \frac{2\sqrt{(-7)^2 - (8)(6)}}{8 + 6}$$

$$\Rightarrow \tan\varphi = \frac{2\sqrt{49-48}}{14} = 1/7$$

$$\therefore \varphi = \tan^{-1}(1/7)$$

Ans

We know that the equⁿ of the bisectors of the angles between the lines is given by

$$\frac{(x-\alpha)^2-(y-\beta)^2}{(x-\alpha)(y-\beta)} = \frac{a-b}{h}$$

$$\Rightarrow \frac{(x-5/2)^2-(y-3)^2}{(x-5/2)(y-3)} = \frac{8-6}{-7}$$

$$\Rightarrow -7\left\{x^2-5x+\frac{25}{4}-y^2+6y-9\right\} = 2(x-5/2)(y-3)$$

$$\Rightarrow -7x^2+35x-\frac{175}{4}+7y^2-42y+63 = (2x-5)(y-3)$$

$$\Rightarrow -7x^2+7y^2+35x-42y+\frac{77}{4} = 2xy-5y-6x+15$$

$$\Rightarrow 7x^2-7y^2-35x+42y-\frac{77}{4}+2xy-5y-6x+15=0$$

$$\Rightarrow 7x^2+2xy-7y^2-41x+37y-\frac{17}{4}=0$$

Ans

From ① we have,

$$8x^2 + (2-14y)x + (6y^2 - y - 1) = 0$$

$$\therefore x = \frac{-(2-14y) \pm \sqrt{(2-14y)^2 - 4.8.(6y^2-y-1)}}{16}$$

$$= \frac{(14y-2) \pm \sqrt{(14y-2)^2 - 32(6y^2-y-1)}}{16}$$

$$= \frac{2(7y-1) \pm \sqrt{196y^2 - 56y + 4 - 192y^2 + 32y + 32}}{16}$$

$$= \frac{2(7y-1) \pm \sqrt{4y^2 - 24y + 36}}{16}$$

$$= \frac{2(7y-1) \pm 2\sqrt{y^2 - 6y + 9}}{16}$$

$$= \frac{7y-1 \pm \sqrt{(y-3)^2}}{8}$$

$$\therefore x = \frac{7y-1 \pm (y-3)}{8}$$

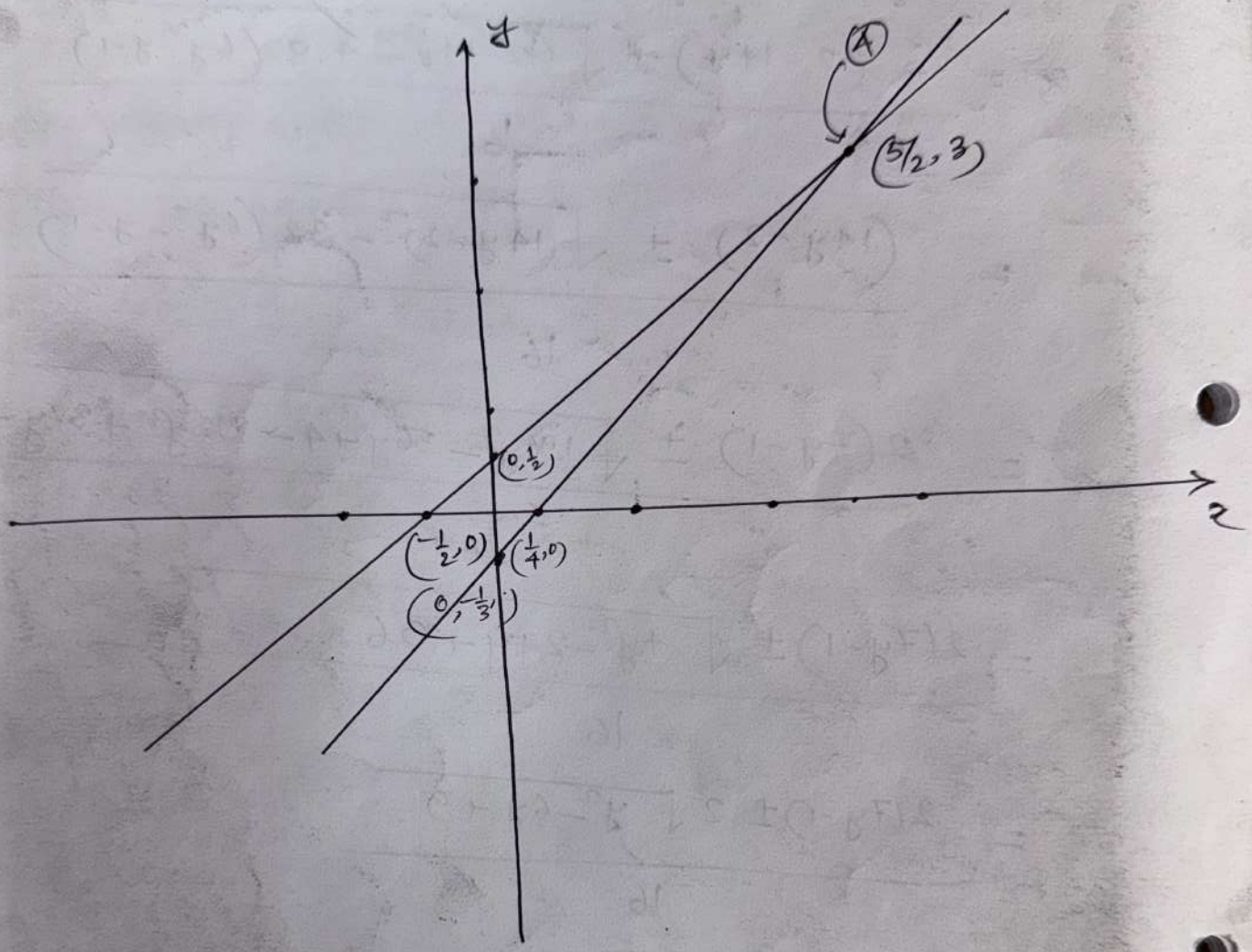
Thus, Either,

$$x = \frac{7y-1+(y-3)}{8} \quad \text{or,} \quad x = \frac{7y-1-(y-3)}{8}$$

$$\Rightarrow 8x = 8y - 4 \qquad \Rightarrow 8x = 6y + 2$$

$$\Rightarrow 2x - 2y + 1 = 0 \text{ ——④} \qquad \Rightarrow 4x - 3y - 1 = 0 \text{ ——⑤}$$

Thus, eqn ④ & ⑤ are the pair of straight lines represented by ①



Problem:

i) $x^2 - xy - 2y^2 - x - 4y - 2 = 0$
ii) $x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$ } H/W

Show that the equation $6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$ represents a pair of straight lines. Also, find the equations of the lines, their point of intersection, angle between the lines, and the eqn of the bisectors of the angles between them.

Soln: Given,

$$6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0 \quad \text{——— ①}$$

Pair of Straight Lines (2)

Comparing equⁿ ① with the general equⁿ of second degree we obtain,

$$a = 6;\ h = -5/2;\ b = -6;\ g = 7,\ f = 5/2;\ c = 4$$

Now,

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

$$= abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= 6(-6)(4) + 2(5/2)(7)(-5/2) - 6(5/2)^2 - (-6)(7)^2 - 4(-5/2)^2$$

$$= -144 - \frac{175}{2} - \frac{75}{2} + 294 + 25$$

$$= 150 - 150$$

$$\therefore \Delta = 0$$

Since, $\Delta = 0$, so equⁿ ① represents a pair of straight lines. And

From ① we can write,

$$6x^2 + (14 - 5y)x - 6y^2 + 5y + 4 = 0$$

$$\Rightarrow 6x^2 + (14 - 5y)x - (6y^2 - 5y - 4) = 0$$

$$\therefore x = \frac{-(14 - 5y) \pm \sqrt{(14 - 5y)^2 + 4.6.(6y^2 - 5y - 4)}}{12}$$

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$$\Rightarrow x = \frac{5y-14 \pm \sqrt{25y^2-140y+196+144y^2-120y}}{12}$$

$$= \frac{5y-14 \pm \sqrt{169y^2-260y+100}}{12}$$

$$= \frac{5y-14 \pm \sqrt{(13y)^2-2\cdot 13y\cdot 10+(10)^2}}{12}$$

$$= \frac{5y-14 \pm \sqrt{(13y-10)^2}}{12}$$

$$\Rightarrow x = \frac{5y-14 \pm (13y-10)}{12}$$

Either, $x = \frac{5y-14+13y-10}{12}$

$\Rightarrow 12x = 18y-24$

$\Rightarrow 2x-3y+4=0$ ——— (2)

or, $x = \frac{5y-14-(13y-10)}{12}$

$\Rightarrow 12x = -8y-4$

$\Rightarrow 3x+2y+1=0$ ——— (3)

Thus, (2) & (3) are the eqn^{n}s of the two straight lines represented by (1).

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Let (α, β) be the point of intersection of ② & ③
Also, assume that

$$F(\alpha,\beta) = 6\alpha^2 - 5\alpha\beta - 6\beta^2 + 14\alpha + 5\beta + 4$$

$$\therefore \frac{\partial F}{\partial \alpha} = 12\alpha - 5\beta + 14 = 0 \quad \text{——— ④}$$

$$\& \quad \frac{\partial F}{\partial \beta} = -5\alpha - 12\beta + 5 = 0 \quad \text{——— ⑤}$$

Solving ④ & ⑤ we obtain

$$\alpha = -\frac{11}{13} \quad \& \quad \beta = \frac{10}{13}$$

Thus, $\left(-\frac{11}{13}, \frac{10}{13}\right)$ is the point of intersection of ② & ③.

Ans

Alternatively, if we solve equⁿ's ② & ③ we obtain

$$x = -\frac{11}{13} \quad \& \quad y = \frac{10}{13}$$

So, $\left(-\frac{11}{13}, \frac{10}{13}\right)$ is the point of intersection of ② & ③.

Let ϕ be the angle between the lines represented by ② & ③, then

$$\tan\phi = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\Rightarrow \tan\phi = \frac{2\sqrt{(-5/2)^2 - 6(-6)}}{6 - 6} = \infty = \tan \pi/2$$

$$\therefore \phi = \pi/2$$

Ans

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The equation of the bisectors of the angles between the lines is given by

$$\frac{(x-\alpha)^2-(y-\beta)^2}{(x-\alpha)(y-\beta)} = \frac{a-b}{h}$$

$$\Rightarrow \frac{\left(x+\frac{11}{13}\right)^2-\left(y-\frac{10}{10}\right)^2}{\left(x+\frac{11}{13}\right)\left(y-\frac{10}{13}\right)} = \frac{6+6}{-5/2}$$

$$\Rightarrow x^2+\frac{22}{13}x+\frac{121}{169}-y^2+\frac{20}{13}y-\frac{100}{169} = -\frac{24}{5}\left\{xy+\frac{11}{13}y - \frac{10}{13}x-\frac{100}{169}\right\}$$

$$\Rightarrow x^2+\frac{22}{13}x-y^2+\frac{20}{13}y+\frac{21}{169} = -\frac{24}{5}xy-\frac{264}{65}y + \frac{240}{65}x+\frac{528}{169}$$

$$\Rightarrow x^2+\frac{24}{5}xy-y^2+\left(\frac{22}{13}-\frac{240}{65}\right)x+\left(\frac{20}{13}+\frac{264}{65}\right)y + \left(\frac{21-528}{169}\right)=0$$

$$\Rightarrow x^2+\frac{24}{5}xy-y^2-2x+\frac{28}{5}y-3=0$$

$$\Rightarrow 5x^2+24xy-5y^2-10x+28y-15=0$$

(Ans)

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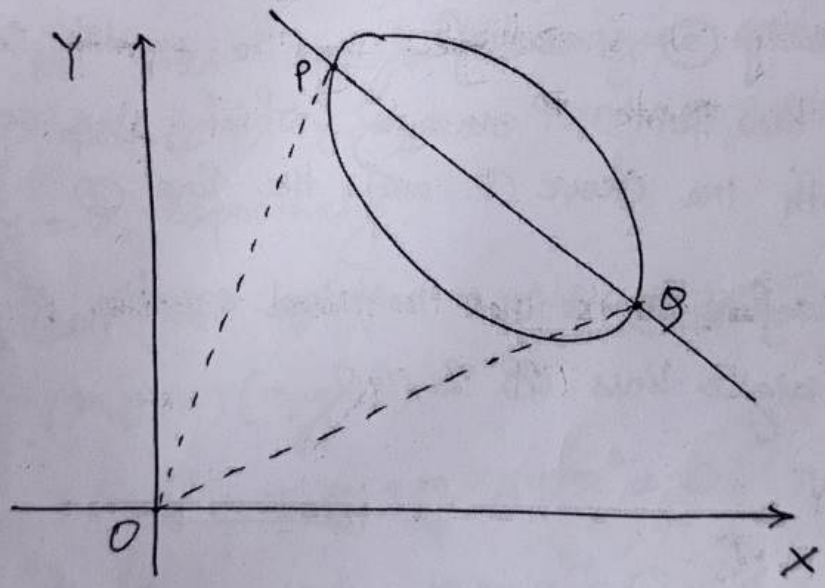
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Problem:

Find the equation of the pair of straight lines joining the origin to the bisections of the curve $ax^2+2hxy+by^2+2gx+2fy+c=0$ and the line $lx+my+n=0$

Soln:

Let the lines intersect the curves at points P and Q. Let O be the origin. Now we are required to find the equation of the line-pair OP & OQ.



We know that a homogeneous equation of the second degree in x, y represents a pair of straight lines that pass through the origin. Hence, the joint equation of OP and OQ must be of the second degree and homogeneous equation in x, y.

Let us now make the equation

$$ax^2+2hxy+by^2+2gx+2fy+c=0 \quad \text{——— (1)}$$

homogeneous (and of the second degree) with respe... to the equation/line $lx+my+n=0$

$$\Rightarrow -\frac{lx+my}{n} = 1 \quad \text{——— (2)}$$

Therefore, we can rewrite equ^n (1) as

$$ax^2+2hxy+by^2+2gx\left(-\frac{lx+my}{n}\right)+2fy\left(-\frac{lx+my}{n}\right)$$

$$+c\left(-\frac{lx+my}{n}\right)^2 = 0 \quad \text{——— (3)}$$

Clearly (3) is satisfied by the ~~conditi~~ coordinates of the points P and Q which are common to both the curve (1) and the line (2).

Therefore, (3) is the required equation of the pair of straight lines OP & OQ.

———o———

Problem:

Prove that the pair of lines $bx^2-2hxy+ay^2=0$ are the perpendiculars through the origin to the pair of lines $ax^2+2hxy+by^2=0$.

Sol^n:

Let $y-m_1x=0$ & $y-m_2x=0$ be the lines that are represented by the equ^n $ax^2+2hxy+by^2=0$

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Therefore,

$$ax^2+2hxy+by^2 \equiv (y-m_1x)(y-m_2x)$$

$$\Rightarrow \frac{a}{b}x^2+\frac{2h}{b}xy+y^2 \equiv y^2-(m_1+m_2)x+m_1m_2x^2$$

Equating the coefficients of the like terms from both sides we have

$$\left.\begin{aligned} m_1+m_2 &= -\frac{2h}{b} \\ m_1m_2 &= a/b \end{aligned}\right\} \quad (1)$$

Now, the lines through the origin and perpendicular to $y-m_1x=0$ & $y-m_2x=0$ are $m_1y+x=0$ and $m_2x+y=0$, respectively.

Hence, their joint equation is given by

$$(m_1y+x)(m_2x+y)=0$$

$$\Rightarrow m_1m_2y^2+m_2xy+m_1xy+x^2=0$$

$$\Rightarrow x^2+(m_1+m_2)xy+m_1m_2y^2=0$$

$$\Rightarrow x^2-\frac{2h}{b}xy+\frac{a}{b}y^2=0$$

$$\Rightarrow bx^2-2hxy+ay^2=0$$

(Proved)

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Problem:
Find the condition that the lines joining the origin to the intersections $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $y=mx+c$ may be coincident.

Sol^n:
Given, the equations are

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad \text{——— (1)}$$

& $y=mx+c \Rightarrow \frac{y-mx}{c}=1$ ——— (2)

Now, make eqn (1) homogeneous with the help of eqn (2)

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=\left(\frac{y-mx}{c}\right)^2$$

$$\Rightarrow \frac{b^2x^2+a^2y^2}{a^2b^2}=\frac{y^2-2mxy+m^2x^2}{c^2}$$

$$\Rightarrow b^2c^2x^2+a^2c^2y^2=a^2b^2y^2-2a^2b^2mxy+a^2b^2m^2x^2$$

$$\Rightarrow b^2(c^2-a^2m^2)x^2+2a^2b^2mxy+a^2(c^2-b^2)y^2=0$$

$$\Rightarrow a'x^2+2h'xy+b'y^2=0 \quad \text{——— (3)}$$

where,
$$a'=b^2(c^2-a^2m^2)$$
$$h'=a^2b^2m$$
$$b'=a^2(c^2-b^2)$$

The lines in equ^n (3) will be coincident if

$$h^2 = a'b'$$

$$\Rightarrow (a^2bm)^2 = b^2(c^2 - a^2m^2)\, a^2(c^2 - b^2)$$

$$\Rightarrow a^4b^4m^2 = a^2b^2(c^2 - a^2m^2)(c^2 - b^2)$$

$$\Rightarrow a^4b^4m^2 = a^2b^2\left(c^4 - a^2c^2m^2 - b^2c^2 + a^2b^2m^2\right)$$

$$\Rightarrow a^4b^4m^2 = a^2b^2c^4 - a^4b^2c^2m^2 - a^2b^4c^2 + a^4b^4m^2$$

$$\Rightarrow a^2b^2c^4 - a^4b^2c^2m^2 - a^2b^4c^2 = 0.$$

$$\Rightarrow a^2b^2c^2(c^2 - a^2m^2 - b^2) = 0$$

$$\Rightarrow c^2 - a^2m^2 - b^2 = 0 \quad [\because a^2b^2c^2 \neq 0]$$

$$\Rightarrow c^2 = a^2m^2 + b^2$$

which is the required condition.

22/11/22

Problem:

Find the equation of the pair of straight lines through the origin perpendicular to the pair given by

$$2x^2 + 5xy + 2y^2 + 10x + 5y = 0$$

Soln:

Given, $2x^2 + 5xy + 2y^2 + 10x + 5y = 0$ ——— (1)

Let $l_1x + m_1y + n_1 = 0$ ——— (2) and $l_2x + m_2y + n_2 = 0$ ——— (3)

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be the straight lines represented by ①, then

$$2x^2+5xy+2y^2+10x+5y \equiv (l_1x+m_1y+n_1)(l_2x+m_2y+n_2)$$

$$\Rightarrow 2x^2+5xy+2y^2+10x+5y \equiv l_1l_2x^2 + l_2m_1xy + l_2n_1x + l_1m_2xy + m_1m_2y^2 + m_2n_1y + l_1n_2x + m_1n_2y + n_1n_2$$

$$\Rightarrow 2x^2+5xy+2y^2+10x+5y \equiv l_1l_2x^2 + (l_1m_2+l_2m_1)xy + m_1m_2y^2 + (l_1n_2+l_2n_1)x + (m_1n_2+m_2n_1)y + n_1n_2$$

Equating the coefficients of the like terms we have

$$l_1l_2=2,\ m_1m_2=2,\ l_1m_2+l_2m_1=5 \quad \text{——— (A)}$$

Now let the straight lines passing through the origin and perpendicular to ② & ③ be

$$m_1x-l_1y=0 \text{ ——— ④} \quad \& \quad m_2x-l_2y=0 \text{ ——— ⑤}$$

respectively

The joint equⁿ of ④ & ⑤ is given by

$$(m_1x-l_1y)(m_2x-l_2y)=0$$

$$\Rightarrow m_1m_2x^2 - l_1m_2xy - l_2m_1xy + l_1l_2y^2 = 0$$

$$\Rightarrow m_1m_2x^2 - (l_1m_2+l_2m_1)xy + l_1l_2y^2 = 0$$

$$\Rightarrow 2x^2 - 5xy + 2y^2 = 0 \quad [\text{using (A)}]$$

And

Quiz 1

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Problem:

Prove that the lines joining the origin to the points of intersections of $ax^2+2hxy+by^2+2gx+2fy+c=0$ and $lx+my+n=0$ are mutually perpendicular if $n^2(a+b)-2n(gl+fm)+c(l^2+m^2)=0$

Sol^n:

The given equations are

$$ax^2+2hxy+by^2+2gx+2fy+c=0 \quad \text{---(1)}$$

& $lx+my+n=0 \Rightarrow -\frac{lx+my}{n}=1$ ---(2)

To find the equ^n of the pair of straight lines through the origin and to the points of intersections of (1) & (2), we have

$$ax^2+2hxy+by^2+2gx\left\{-\left(\frac{lx+my}{n}\right)\right\}+2fy\left\{-\left(\frac{lx+my}{n}\right)\right\}+c\left\{-\left(\frac{lx+my}{n}\right)\right\}^2=0$$

$$\Rightarrow ax^2+2hxy+by^2-\frac{2gx(lx+my)}{n}-\frac{2fy(lx+my)}{n}+\frac{c(lx+my)^2}{n^2}=0$$

$$\Rightarrow an^2x^2+2hn^2xy+bn^2y^2-2gxn(lx+my)-2fyn(lx+my)+c(lx+my)^2=0$$

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$$\Rightarrow an^2x^2 + 2hn^2xy + bn^2y^2 - 2gln x^2 - 2gmnxy - 2flnxy - 2fmny^2 + cl^2x^2 + 2clmxy + cm^2y^2 = 0$$

$$\Rightarrow (an^2 - 2gln + cl^2)x^2 + (2hn^2 - 2gmn - 2fln + 2clm)xy + (bn^2 - 2fmn + cm^2)y^2 = 0$$

$$\Rightarrow a'x^2 + 2h'xy + b'y^2 = 0 \quad \text{---------- (3)}$$

where,

$$a' = an^2 - 2gln + cl^2$$

$$2h' = 2hn^2 - 2gmn - 2fln + 2clm$$

$$\Rightarrow h' = hn^2 - gmn - fln + clm$$

$$\& \quad b' = bn^2 - 2fmn + cm^2$$

} ---------- (4)

The pair of straight lines represented by (3) will be perpendicular if

$$a' + b' = 0$$

$$\Rightarrow an^2 - 2gln + cl^2 + bn^2 - 2fmn + cm^2 = 0 \quad [\text{using (4)}]$$

$$\Rightarrow n^2(a+b) - 2n(gl + fm) + c(l^2 + m^2) = 0$$

(Showed)

Problem:

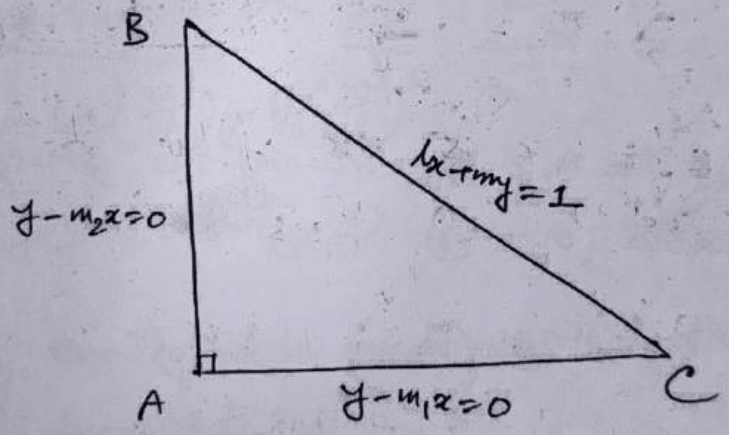
Show that the triangle formed by the lines $ax^2 + 2hxy + by^2 = 0$ and $lx + my = 1$ is right angled if $(a+b)(al^2 + 2hlm + bm^2) = 0$.

Soln:

Given the lines are

$$ax^2 + 2hxy + by^2 = 0 \quad \text{---} \quad (1)$$

$$\& \; lx + my = 1 \quad \text{---} \quad (2)$$



Let the lines represented by (1) are $y - m_1x = 0$ & $y - m_2x = 0$, respectively.

Therefore, we can write

$$ax^2 + 2hxy + by^2 \equiv (y - m_1x)(y - m_2x)$$

$$\Rightarrow y^2 + \frac{2h}{b}xy + \frac{a}{b}x^2 \equiv y^2 - (m_1 + m_2)x + m_1m_2x^2$$

Equating the coefficients of the like terms from both

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sides we have

$$\left.\begin{aligned} m_1 + m_2 &= -\frac{2h}{b} \\ m_1 m_2 &= \frac{a}{b} \end{aligned}\right\} \quad \text{——— (3)}$$

From the figure, if $AB \perp AC$, i.e. if $\angle A = 90°$, then

$$a + b = 0 \quad \text{——— (A)}$$

If $AB \perp BC$, i.e., if $\angle B = 90°$, then

$$m_2\left(-\frac{l}{m}\right) = -1$$

$$\Rightarrow m_2 = \frac{m}{l} \quad \text{——— (4)}$$

Since, $y = m_2 x$ is one of the straight lines represented by (1), thus

$$ax^2 + 2hx(m_2 x) + b(m_2 x)^2 = 0$$

$$\Rightarrow ax^2 + 2hm_2 x^2 + bm_2^2 x^2 = 0$$

$$\Rightarrow a + 2hm_2 + bm_2^2 = 0 \quad [\because x^2 \neq 0]$$

$$\Rightarrow a + 2h\left(\frac{m}{l}\right) + b\left(\frac{m}{l}\right)^2 = 0 \quad [\text{using (4)}]$$

$$\Rightarrow a + \frac{2hm}{l} + \frac{bm^2}{l^2} = 0$$

$$\Rightarrow al^2 + 2hlm + bm^2 = 0 \quad \text{——— (B)}$$

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Again if $BC \perp AC$, i.e., if $\angle C = 90^\circ$ then

$$m_1\left(-\frac{l}{m}\right) = -1$$

$$\Rightarrow m_1 = \frac{m}{l}$$

Since, $y = m_1 x$ is one of the straight lines represented by ①, then

$$ax^2 + 2hx(m_1 x) + b(m_1 x)^2 = 0$$

$$\Rightarrow ax^2 + 2hm_1 x^2 + bm_1^2 x^2 = 0$$

$$\Rightarrow a + 2hm_1 + bm_1^2 = 0 \quad [\because x^2 \neq 0]$$

$$\Rightarrow a + 2h\left(\frac{m}{l}\right) + b\left(\frac{m}{l}\right)^2 = 0$$

$$\Rightarrow al^2 + 2hml + bm^2 = 0 \quad \text{——— (C)}$$

Therefore, the required conditions that the triangle ABC will be right angled are given by

$$(a+b)(al^2 + 2hlm + bm^2) = 0 \quad [\text{using (A), (B), (C)}]$$

(Showed)

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Problem:

If the equation $ax^2+2hxy+by^2+2gx+2fy+c=0$ represents a pair of straight lines equidistant from the origin, then prove that

(i) $f^2g^2 = c(bf^2-ag^2)$

(ii) $h(g^2-f^2) = fg(a-b)$.

Soln: i)

Given,

$$ax^2+2hxy+by^2+2gx+2fy+c=0 \quad \text{——— (1)}$$

Let the two straight lines represented by (1) are

$$l_1x+m_1y+n_1=0 \quad \text{——— (2)}$$

& $$l_2x+m_2y+n_2=0 \quad \text{——— (3)}$$

Therefore, we can write

$$ax^2+2hxy+by^2+2gx+2fy+c \equiv (l_1x+m_1y+n_1)(l_2x+m_2y+n_2)$$

$$\Rightarrow ax^2+2hxy+by^2+2gx+2fy+c \equiv l_1l_2x^2+l_2m_1xy+l_2n_1x + l_1m_2xy+m_1m_2y^2+m_2n_1y + l_1m_2x+m_1n_2y+n_1n_2$$

Equating the coefficients of the like terms from both sides we have,

$l_1 l_2 = a$; $l_1 m_2 + l_2 m_1 = 2h$; $m_1 m_2 = b$;

$l_1 n_2 + l_2 n_1 = 2g$; $m_1 n_2 + m_2 n_1 = 2f$; $n_1 n_2 = c$;

Given that lines ② & ③ are equidistant from the origin, then we have

$$\left|\frac{n_1}{\sqrt{l_1^2+m_1^2}}\right| = \left|\frac{n_2}{\sqrt{l_2^2+m_2^2}}\right|$$

$$\Rightarrow \frac{n_1^2}{l_1^2+m_1^2} = \frac{n_2^2}{l_2^2+m_2^2}$$

$$\Rightarrow n_1^2(l_2^2+m_2^2) = n_2^2(l_1^2+m_1^2)$$

$$\Rightarrow n_1^2 l_2^2 + n_1^2 m_2^2 = n_2^2 l_1^2 + n_2^2 m_1^2$$

$$\Rightarrow n_1^2 l_2^2 - n_2^2 l_1^2 = n_2^2 m_1^2 - n_1^2 m_2^2$$

$$\Rightarrow (n_1 l_2 + n_2 l_1)(n_1 l_2 - n_2 l_1) = (n_2 m_1 + n_1 m_2)(n_2 m_1 - n_1 m_2)$$

$$\Rightarrow (l_1 n_2 + l_2 n_1)(l_2 n_1 - l_1 n_2) = (m_1 n_2 + m_2 n_1)(n_2 m_1 - n_1 m_2)$$

$$\Rightarrow 2g\sqrt{(l_2 n_1 + l_1 n_2)^2 - 4 l_1 l_2 n_1 n_2} = 2f\sqrt{(m_1 n_2 + m_2 n_1)^2 - 4 m_1 m_2 n_1 n_2}$$

$$\Rightarrow 2g\sqrt{(2g)^2 - 4ac} = 2f\sqrt{(2f)^2 - 4bc}$$

$$\Rightarrow g\sqrt{4g^2 - 4ac} = f\sqrt{4f^2 - 4bc}$$

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$\Rightarrow g\sqrt{g^2-ac} = f\sqrt{f^2-bc}$

$\Rightarrow g^2(g^2-ac) = f^2(f^2-bc)$

$\Rightarrow g^4 - g^2ac = f^4 - f^2bc$

$\Rightarrow f^4 - g^4 = f^2bc - g^2ac$

$\Rightarrow f^4 - g^4 = c(bf^2 - ag^2)$ ——— (4)

(Proved)

ii) Since equⁿ (1) represents a pair of straight line, so $\Delta = 0$

$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$\Rightarrow (ab-h^2)c = af^2 + bg^2 - 2fgh$

$\Rightarrow c = \dfrac{af^2+bg^2-2fgh}{ab-h^2}$

$\Rightarrow \dfrac{f^4-g^4}{bf^2-ag^2} = \dfrac{af^2+bg^2-2fgh}{ab-h^2}$ [using (4)]

$\therefore$

$\Rightarrow (f^4-g^4)(ab-h^2) = (bf^2-ag^2)(af^2+bg^2-2fgh)$

$\Rightarrow abf^4 - abg^4 - h^2f^4 + g^4h^2 = abf^4 - a^2f^2g^2 + b^2f^2g^2 - abg^4 - 2bf^3gh + 2afg^3h$

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$$\Rightarrow a^2f^2g^2 - 2afg^3h + g^4h^2 = b^2f^2g^2 - 2bf^3gh + h^2f^4$$

$$\Rightarrow (afg)^2 - 2fg.(hg^2) + (hg^2)^2 = (bfg)^2 - 2bfg(hf^2) + (hf^2)^2$$

$$\Rightarrow (afg - hg^2)^2 = (bfg - hf^2)^2$$

$$\Rightarrow afg - hg^2 = bfg - hf^2$$

$$\Rightarrow afg - bfg = hg^2 - hf^2$$

$$\Rightarrow h(g^2 - f^2) = fg(a - b)$$

(Proved)

Problem:

Show that one of the two lines represented by $ax^2 + 2hxy + by^2 = 0$ will coincide with one of the two lines given by $a_1x^2 + 2h_1xy + b_1y^2 = 0$ if

$$(ab_1 - a_1b)^2 = 4(a_1h - ah_1)(bh_1 - b_1h)$$

Sol^n:

Given, $ax^2 + 2hxy + by^2 = 0$ ——— (1)

& $a_1x^2 + 2h_1xy + b_1y^2 = 0$ ——— (2)

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Let the common straight line of ① & ② be

$$y = mx$$

From ① we can write

$$ax^2 + 2hx^2m + bx^2m^2 = 0$$

$$\Rightarrow bm^2 + 2hm + a = 0 \quad [\because x^2 \neq 0] \quad \text{——— ③}$$

Similarly, from ② we can write

$$b_1m^2 + 2h_1m + a_1 = 0 \quad \text{——— ④}$$

Solving ③ & ④ we have

$$\frac{m^2}{2ha_1 - 2h_1a} = \frac{m}{ab_1 - a_1b} = \frac{1}{2h_1b - 2hb_1}$$

$$m = \frac{2(a_1h - ah_1)}{ab_1 - a_1b} \quad \& \quad m = \frac{ab_1 - a_1b}{2(h_1b - b_1h)}$$

Problem:

Prove that two of the straight lines represented by the equⁿ $ax^3 + bx^2y + cxy^2 + dy^3 = 0$, will be at right angles if $a^2 + ac + bd + d^2 = 0$

Solⁿ:

Given,

$$ax^3 + bx^2y + cxy^2 + dy^3 = 0 \quad \text{——— ①}$$

Here, equⁿ ① is a homogeneous equⁿ of degree-3. So, the given equⁿ will represent three straight lines that pass through the origin.

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Let the equⁿ of the pair of straight lines perpendicular to each other be

$$x^2 + kxy - y^2 = 0 \quad \text{——— (2)}$$

Perpendicular SLs.
$a_1x + b_1y = 0$
$b_1x - a_1y = 0$

& the equⁿ of the straight line be

$$ax - dy = 0 \quad \text{——— (3)}$$

Thus,

$$ax^3 + bx^2y + cxy^2 + dy^3 \equiv (x^2 + kxy - y^2)(ax - dy)$$

$$= ax^3 + akx^2y - axy^2 - dx^2y - dkxy^2 + dy^3$$

$$\Rightarrow ax^3 + bx^2y + cxy^2 + dy^3 = ax^3 + x^2y(ak - d) - xy^2(a + dk) + dy^3$$

Equating the coefficients of x^2y & xy^2 from both sides we get

$$b = ak - d \quad \& \quad c = -(a + dk)$$

$$\Rightarrow k = \frac{b+d}{a} \quad \Rightarrow c = -a - d\left(\frac{b+d}{a}\right)$$

$$\Rightarrow c = -a - \frac{bd + d^2}{a}$$

$$\Rightarrow c + a + \frac{bd + d^2}{a} = 0$$

$$\Rightarrow ac + a^2 + bd + d^2 = 0$$

$$\Rightarrow a^2 + ac + bd + d^2 = 0$$

(Proved)

Lecture 16
04.11.22

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Example:

If $ax^2+2hxy+by^2+2gx+2fy+c=0$ represents a pair of straight lines, then show that the area of the triangle formed by their bisectors and x-axis is

$$\frac{\sqrt{\{(a-b)^2+4h^2\}}}{2h}\cdot\frac{(ca-g^2)}{(ab-h^2)}$$

Sol^n:

Given, $ax^2+2hxy+by^2+2gx+2fy+c=0$ ——— (1)

represents a pair of straight lines if

$$\Delta=\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}=0$$

$\Rightarrow abc+2fgh-af^2-bg^2-ch^2=0$ ——— (2)

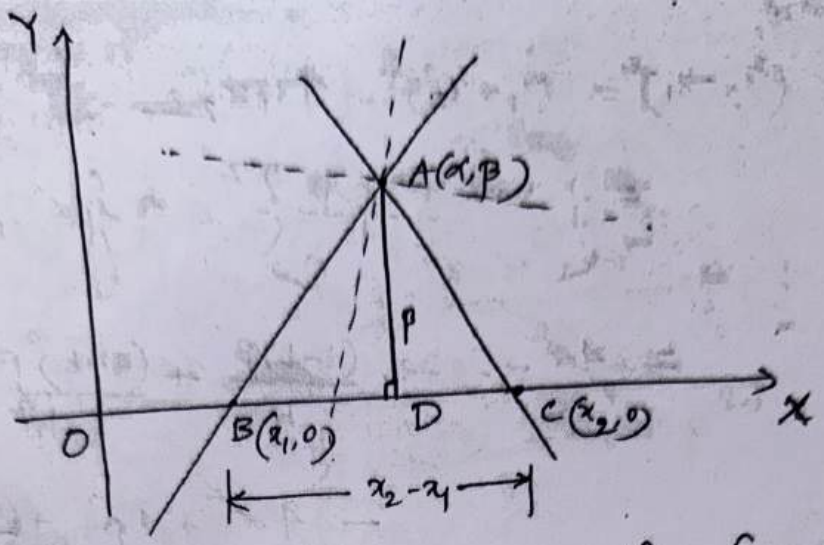
Let (α,β) be the point of intersection of the straight lines represented by (1), then

$$\alpha=\frac{hf-bg}{ab-h^2} \quad \& \quad \beta=\frac{hg-af}{ab-h^2}$$

We know, that the eqn^n of bisectors

$$\frac{(x-\alpha)^2-(y-\beta)^2}{(x-\alpha)(y-\beta)}=\frac{a-b}{h} \quad ——— (3)$$

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The equⁿ of X-axis is $y = 0$, then from ③ we get

$$\frac{(x-\alpha)^2-\beta^2}{(x-\alpha)(-\beta)} = \frac{a-b}{h}$$

$$\Rightarrow h\{x^2-2x\alpha+\alpha^2-\beta^2\} = -\beta(x-\alpha)(a-b)$$

$$\Rightarrow hx^2-2x\alpha h+h(\alpha^2-\beta^2)+\beta(a-b)x-\alpha\beta(a-b)=0$$

$$\Rightarrow hx^2-\{2h\alpha-(a-b)\beta\}x+\{h(\alpha^2-\beta^2)-\alpha\beta(a-b)\}=0 \quad \text{---④}$$

which is a quadratic equⁿ of x

Let the roots of equⁿ ④ be x_1 & x_2

$$\therefore x_1+x_2 = \frac{2h\alpha-(a-b)\beta}{h} = 2\alpha-\frac{(a-b)\beta}{h}$$

$$\& \quad x_1x_2 = \frac{h(\alpha^2-\beta^2)-\alpha\beta(a-b)}{h} = \alpha^2-\beta^2-\frac{(a-b)}{h}\alpha\beta$$

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Now,

$$(x_2-x_1)^2 = (x_1+x_2)^2 - 4x_1x_2$$

$$= \left\{2\alpha - \frac{(a-b)\beta}{h}\right\}^2 - 4\left\{(\alpha^2-\beta^2) + \left(\frac{a-b}{h}\right)\alpha\beta\right\}$$

$$= 4\alpha^2 - 2\cdot 2\alpha\cdot\frac{(a-b)\beta}{h} + \frac{(a-b)^2\beta^2}{h^2} - 4\alpha^2 + 4\beta^2 + \frac{4(a-b)}{h}\alpha\beta$$

$$= \frac{(a-b)^2}{h^2}\beta^2 + 4\beta^2 - \frac{4(a-b)\alpha\beta}{h} + \frac{4(a-b)\alpha\beta}{h}$$

$$= \frac{(a-b)^2\beta^2 + 4h^2\beta^2}{h^2}$$

$$= \frac{\{(a-b)^2 + 4h^2\}\beta^2}{h^2}$$

$$\therefore x_2 - x_1 = \frac{\sqrt{(a-b)^2+4h^2}}{h}\cdot\beta$$

From the figure, we can say that the vertices of the triangle ABC are $A(\alpha,\beta)$, $B(x_1,0)$ & $C(x_2,0)$. Also, the height of the ΔABC on BC is β.

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$\therefore$ Area of $\Delta ABC = \frac{1}{2} \cdot BC \cdot AD$

$$= \frac{1}{2} \cdot (a_2 - a_1) \cdot \beta$$

$$= \frac{1}{2} \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \cdot \beta \cdot \beta$$

$$= \frac{1}{2} \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \cdot \beta^2$$

$$= \frac{1}{2} \cdot \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \cdot \left(\frac{hg - af}{ab - h^2} \right)^2$$

$$= \frac{1}{2} \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \left\{ \frac{h^2g^2 - 2hgaf + a^2f^2}{(ab-h^2)^2} \right\}$$

$$= \frac{1}{2} \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \left\{ \frac{h^2g^2 + a(af^2 - 2fgh)}{(ab-h^2)^2} \right\}$$

$$= \frac{1}{2} \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \left\{ \frac{h^2g^2 + a(abc - bg^2 - ch^2)}{(ab-h^2)^2} \right\}$$

[using ②]

$$= \frac{1}{2} \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \left\{ \frac{h^2g^2 + a^2bc - abg^2 - ach^2}{(ab-h^2)^2} \right\}$$

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$$= \frac{1}{2}\frac{\sqrt{(a-b)^2+4h^2}}{h}\left\{\frac{ac(ab-h^2)-g^2(ab-h^2)}{(ab-h^2)^2}\right\}$$

$$= \frac{1}{2}\frac{\sqrt{(a-b)^2+4h^2}}{h}\left\{\frac{(ab-h^2)(ac-g^2)}{(ab-h^2)^2}\right\}$$

$$= \frac{1}{2}\frac{\sqrt{(a-b)^2+4h^2}}{h}\cdot\frac{(ca-g^2)}{(ab-h^2)}$$

(Showed)

Example:

Prove that $x^2+6xy+9y^2+4x+12y-5=0$ represents a pair of parallel straight lines and the distance between them is $\frac{2\sqrt{10}}{5}$.

Sol^n:

Given,

$$x^2+6xy+9y^2+4x+12y-5=0 \quad \text{---(1)}$$

$$\Rightarrow x^2+2(3y+2)x+(9y^2+12y-5)=0$$

$$\therefore x = \frac{-2(3y+2)\pm\sqrt{4(3y+2)^2-4.1(9y^2+12y-5)}}{2.1}$$

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$$\Rightarrow x = -(3y+2) \pm \sqrt{(3y+2)^2 - (9y^2+12y-5)}$$

$$\Rightarrow x = -(3y+2) \pm \sqrt{9y^2+12y+4-9y^2-12y+5}$$

$$\Rightarrow x = -(3y+2) \pm 3$$

$$\therefore x = -3y-2+3 \quad \text{---(II)}$$

$$\Rightarrow x = -3y+1$$

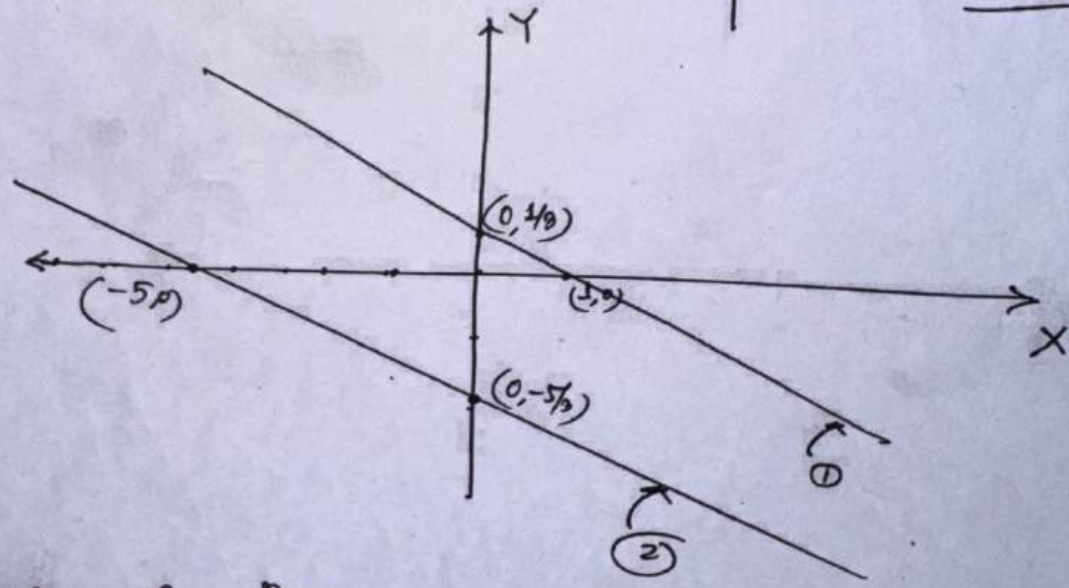
$$\Rightarrow x+3y-1=0 \quad \text{---(II)}$$

OR,

$$x = -3y-2-3$$

$$\Rightarrow x = -3y-5$$

$$\Rightarrow x+3y+5=0 \quad \text{---(III)}$$



Slope of eqn (II), $m_1 = -1/3$

Slope of eqn (III), $m_2 = -1/3$

So, the given eqn (I) represents a pair of parallel straight lines.

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The distance between the two lines (15) & (14) is

$$d = \left| \frac{5-(-1)}{\sqrt{1^2+3^2}} \right| = \frac{6}{\sqrt{10}} = \frac{6\sqrt{10}}{10}$$

$$\Rightarrow d = \frac{3\sqrt{10}}{5}$$

Ans

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Let (α, β) be the point of intersection of ② & ③

Also, assume that

$$F(\alpha,\beta) = 6\alpha^2 - 5\alpha\beta - 6\beta^2 + 14\alpha + 5\beta + 4$$

$$\therefore \frac{\partial F}{\partial \alpha} = 12\alpha - 5\beta + 14 = 0 \quad \text{——— ④}$$

$$\& \quad \frac{\partial F}{\partial \beta} = -5\alpha - 12\beta + 5 = 0 \quad \text{——— ⑤}$$

Solving ④ & ⑤ we obtain

$$\alpha = -\frac{11}{13} \quad \& \quad \beta = \frac{10}{13}$$

Thus, $\left(-\frac{11}{13}, \frac{10}{13}\right)$ is the point of intersection of ② & ③.

~~Ans~~

Alternatively, if we solve equ^{n}s ② & ③ we obtain

$$x = -\frac{11}{13} \quad \& \quad y = \frac{10}{13}$$

So, $\left(-\frac{11}{13}, \frac{10}{13}\right)$ is the point of intersection of ② & ③

Let φ be the angle between the lines represented by ② & ③, then

$$\tan\varphi = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\Rightarrow \tan\varphi = \frac{2\sqrt{(-5/2)^2 - 6(-6)}}{6 - 6} = \infty = \tan \pi/2$$

$$\therefore \varphi = \pi/2$$

~~Ans~~

The equation of the bisectors of the angles between the lines is given by

$$\frac{(x-\alpha)^2-(y-\beta)^2}{(x-\alpha)(y-\beta)} = \frac{a-b}{h}$$

$$\Rightarrow \frac{\left(x+\frac{11}{13}\right)^2-\left(y-\frac{10}{13}\right)^2}{\left(x+\frac{11}{13}\right)\left(y-\frac{10}{13}\right)} = \frac{6+6}{-5/2}$$

$$\Rightarrow x^2+\frac{22}{13}x+\frac{121}{169}-y^2+\frac{20}{13}y-\frac{100}{169} = -\frac{24}{5}\left\{xy+\frac{11}{13}y-\frac{10}{13}x-\frac{100}{169}\right\}$$

$$\Rightarrow x^2+\frac{22}{13}x-y^2+\frac{20}{13}y+\frac{21}{169} = -\frac{24}{5}xy-\frac{264}{65}y+\frac{240}{65}x+\frac{528}{169}$$

$$\Rightarrow x^2+\frac{24}{5}xy-y^2+\left(\frac{22}{13}-\frac{240}{65}\right)x+\left(\frac{20}{13}+\frac{264}{65}\right)y+\left(\frac{21-528}{169}\right)=0$$

$$\Rightarrow x^2+\frac{24}{5}xy-y^2-2x+\frac{28}{5}y-3=0$$

$$\Rightarrow 5x^2+24xy-5y^2-10x+28y-15=0$$

(Ans)

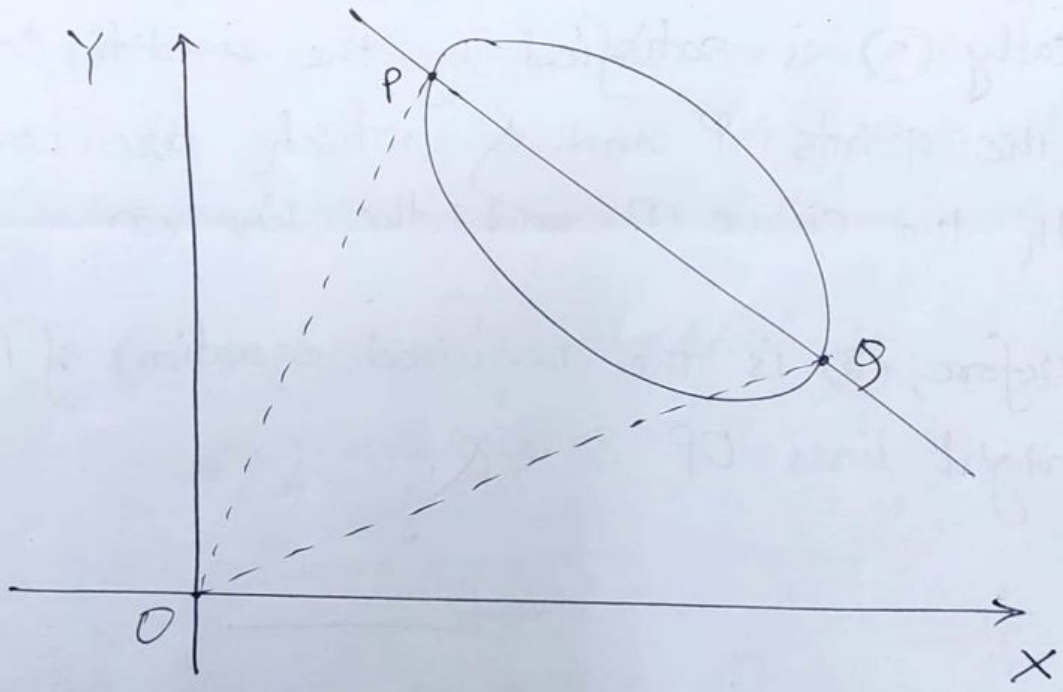
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Problem:

Find the equation of the pair of straight lines joining the origin to the bisections of the curve $ax^2+2hxy+by^2+2gx+2fy+c=0$ and the line $lx+my+n=0$

Soln:

Let the lines intersect the curves at points P and Q. Let O be the origin. Now we are required to find the equation of the line-pair OP & OQ.



We know that a homogeneous equation of the second degree in x, y represents a pair of straight lines that pass through the origin. Hence, the joint equation of OP and OQ must be of the second degree and homogeneous equation in x, y.

Let us now make the equation

$$ax^2+2hxy+by^2+2gx+2fy+c=0 \quad \text{——— (1)}$$

homogeneous (and of the second degree) with respect to the equation/line $lx+my+n=0$

$$\Rightarrow -\frac{lx+my}{n} = 1 \quad \text{——— (2)}$$

Therefore, we can rewrite equ^n (1) as

$$ax^2+2hxy+by^2+2gx\left(-\frac{lx+my}{n}\right)+2fy\left(-\frac{lx+my}{n}\right)$$

$$+c\left(-\frac{lx+my}{n}\right)^2 = 0 \quad \text{——— (3)}$$

Clearly (3) is satisfied by the ~~conditi~~ coordinates of the points P and Q which are common to both the curve (1) and the line (2).

Therefore, (3) is the required equation of the pair of straight lines OP & OQ.

Problem:

Prove that the pair of lines $bx^2-2hxy+ay^2=0$ are the perpendiculars through the origin to the pair of lines $ax^2+2hxy+by^2=0$.

Sol^n:

Let $y-m_1x=0$ & $y-m_2x=0$ be the lines that are represented by the equ^n $ax^2+2hxy+by^2=0$

Therefore,

$$ax^2 + 2hxy + by^2 \equiv (y - m_1x)(y - m_2x)$$

$$\Rightarrow \frac{a}{b}x^2 + \frac{2h}{b}xy + y^2 \equiv y^2 - (m_1 + m_2)x + m_1m_2x^2 \quad \text{---}$$

Equating the coefficients of the like terms from both sides we have

$$\left.\begin{aligned} m_1 + m_2 &= -\frac{2h}{b} \\ m_1m_2 &= a/b \end{aligned}\right\} \quad \text{---} (1)$$

Now, the lines through the origin and perpendicular to $y - m_1x = 0$ & $y - m_2x = 0$ are $m_1y + x = 0$ and $m_2x + y = 0$, respectively.

Hence, their joint equation is given by

$$(m_1y + x)(m_2x + y) = 0$$

$$\Rightarrow m_1m_2y^2 + m_2xy + m_1xy + x^2 = 0$$

$$\Rightarrow x^2 + (m_1 + m_2)xy + m_1m_2y^2 = 0$$

$$\Rightarrow x^2 - \frac{2h}{b}xy + \frac{a}{b}y^2 = 0$$

$$\Rightarrow bx^2 - 2hxy + ay^2 = 0$$

(Proved)

Problem:

find the condition that the lines joining the origin to the intersections $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $y = mx + c$ may be coincident.

Soln:

Given, the equations are

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{——— (1)}$$

& $y = mx + c \Rightarrow \frac{y - mx}{c} = 1$ ——— (2)

Now, make eqn (1) homogeneous with the help of eqn (2)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \left(\frac{y - mx}{c}\right)^2$$

$$\Rightarrow \frac{b^2x^2 + a^2y^2}{a^2b^2} = \frac{y^2 - 2mxy + m^2x^2}{c^2}$$

$$\Rightarrow b^2c^2x^2 + a^2c^2y^2 = a^2b^2y^2 - 2a^2b^2mxy + a^2b^2m^2x^2$$

$$\Rightarrow b^2(c^2 - a^2m^2)x^2 + 2a^2b^2mxy + a^2(c^2 - b^2)y^2 = 0$$

$$\Rightarrow a'x^2 + 2h'xy + b'y^2 = 0 \quad \text{——— (3)}$$

where,

$$a' = b^2(c^2 - a^2m^2)$$

$$h' = a^2b^2m$$

$$b' = a^2(c^2 - b^2)$$

The lines in equⁿ (3) will be coincident if

$$h^2 = a'b'$$

$$\Rightarrow (a^2b^2m)^2 = b^2(c^2 - a^2m^2)\,a^2(c^2 - b^2)$$

$$\Rightarrow a^4b^4m^2 = a^2b^2(c^2 - a^2m^2)(c^2 - b^2)$$

$$\Rightarrow a^4b^4m^2 = a^2b^2\left(c^4 - a^2c^2m^2 - b^2c^2 + a^2b^2m^2\right)$$

$$\Rightarrow a^4b^4m^2 = a^2b^2c^4 - a^4b^2c^2m^2 - a^2b^4c^2 + a^4b^4m^2$$

$$\Rightarrow a^2b^2c^4 - a^4b^2c^2m^2 - a^2b^4c^2 = 0.$$

$$\Rightarrow a^2b^2c^2(c^2 - a^2m^2 - b^2) = 0$$

$$\Rightarrow c^2 - a^2m^2 - b^2 = 0 \qquad [\because a^2b^2c^2 \neq 0]$$

$$\Rightarrow c^2 = a^2m^2 + b^2$$

which is the required condition.

Problem:

Find the equation of the pair of straight lines through the origin perpendicular to the pair given by

$$2x^2 + 5xy + 2y^2 + 10x + 5y = 0$$

Solⁿ:

Given, $2x^2 + 5xy + 2y^2 + 10x + 5y = 0$ ——— (1)

Let $l_1x + m_1y + n_1 = 0$ ——— (2) and $l_2x + m_2y + n_2 = 0$ ——— (3)

be the straight lines represented by ①, then

$$2x^2+5xy+2y^2+10x+5y \equiv (l_1x+m_1y+n_1)(l_2x+m_2y+n_2)$$

$$\Rightarrow 2x^2+5xy+2y^2+10x+5y \equiv l_1l_2x^2+l_2m_1xy+l_2n_1x + l_1m_2xy+m_1m_2y^2+m_2n_1y + l_1n_2x+m_1n_2y+n_1n_2$$

$$\Rightarrow 2x^2+5xy+2y^2+10x+5y \equiv l_1l_2x^2+(l_1m_2+l_2m_1)xy + m_1m_2y^2+(l_1n_2+l_2n_1)x + (m_1n_2+m_2n_1)y+n_1n_2$$

Equating the coefficients of the like terms we have

$$l_1l_2=2,\ m_1m_2=2,\ l_1m_2+l_2m_1=5 \quad \text{(A)}$$

Now let the straight lines passing through the origin and perpendicular to ② & ③ be

$$m_1x-l_1y=0 \quad \text{④} \quad \& \quad m_2x-l_2y=0 \quad \text{⑤}$$

respectively

The joint eqn of ④ & ⑤ is given by

$$(m_1x-l_1y)(m_2x-l_2y)=0$$

$$\Rightarrow m_1m_2x^2-l_1m_2xy-l_2m_1xy+l_1l_2y^2=0$$

$$\Rightarrow m_1m_2x^2-(l_1m_2+l_2m_1)xy+l_1l_2y^2=0$$

$$\Rightarrow 2x^2-5xy+2y^2=0 \quad [\text{using (A)}]$$

Ans

Quiz 1

Problem:

Prove that the line joining the origin to the points of intersections of $ax^2+2hxy+by^2+2gx+2fy+c=0$ and $lx+my+n=0$ are mutually perpendicular if $n^2(a+b)-2n(gl+fm)+c(l^2+m^2)=0$

Sol^n:

The given equations are

$$ax^2+2hxy+by^2+2gx+2fy+c=0 \quad \text{---(1)}$$

& $$lx+my+n=0 \Rightarrow -\frac{lx+my}{n}=1 \quad \text{---(2)}$$

To find the equ^n of the pair of straight lines through the origin and to the points of intersections of (1) & (2), we have

$$ax^2+2hxy+by^2+2gx\left\{-\left(\frac{lx+my}{n}\right)\right\}+2fy\left\{-\left(\frac{lx+my}{n}\right)\right\}$$
$$+c\left\{-\left(\frac{lx+my}{n}\right)\right\}^2=0$$

$$\Rightarrow ax^2+2hxy+by^2-\frac{2gx(lx+my)}{n}-\frac{2fy(lx+my)}{n}$$
$$+\frac{c(lx+my)^2}{n^2}=0$$

$$\Rightarrow an^2x^2+2hn^2xy+bn^2y^2-2gxn(lx+my)-2fyn(lx+my)$$
$$+c(lx+my)^2=0$$

$$\Rightarrow an^2x^2 + 2hn^2xy + bn^2y^2 - 2gln x^2 - 2gmnxy - 2flnxy - 2fmny^2 + cl^2x^2 + 2clmxy + cm^2y^2 = 0$$

$$\Rightarrow (an^2 - 2gln + cl^2)x^2 + (2hn^2 - 2gmn - 2fln + 2clm)xy + (bn^2 - 2fmn + cm^2)y^2 = 0$$

$$\Rightarrow a'x^2 + 2h'xy + b'y^2 = 0 \quad \text{---------- (3)}$$

where,

$$\left.\begin{aligned} a' &= an^2 - 2gln + cl^2 \\ 2h' &= 2hn^2 - 2gmn - 2flu + 2clu \\ \Rightarrow h' &= hn^2 - gmn - fln + clu \\ \& \quad b' &= bn^2 - 2fmn + cm^2 \end{aligned}\right\} \text{---------- (4)}$$

The pair of straight lines represented by (3) will be perpendicular if

$$a' + b' = 0$$

$$\Rightarrow an^2 - 2gln + cl^2 + bn^2 - 2fmn + cm^2 = 0 \quad [\text{using (4)}]$$

$$\Rightarrow n^2(a+b) - 2n(gl + fm) + c(l^2 + m^2) = 0$$

(Showed)

Problem:

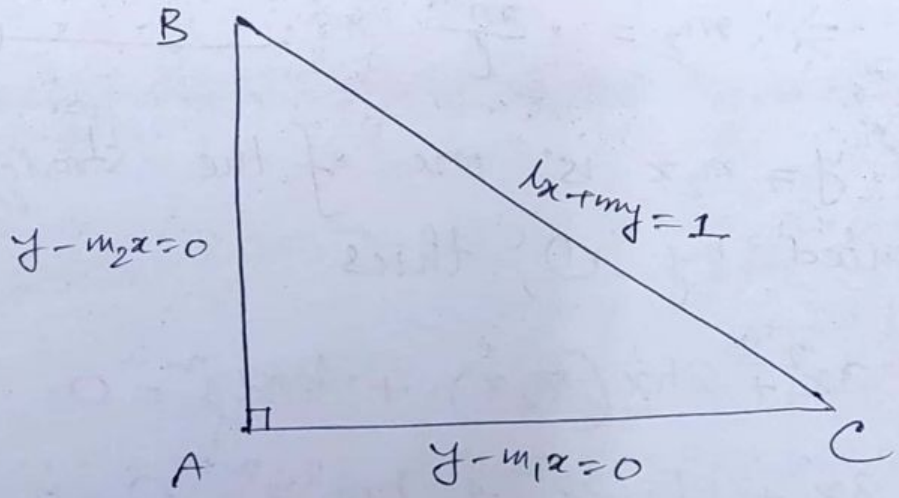
Show that the triangle formed by the lines $ax^2 + 2hxy + by^2 = 0$ and $lx + my = 1$ is right angled if $(a+b)(al^2 + 2hlm + bm^2) = 0$.

Soln:

Given the lines are

$$ax^2 + 2hxy + by^2 = 0 \quad \text{——— (1)}$$

$$\& \quad lx + my = 1 \quad \text{——— (2)}$$



Let the lines represented by (1) are $y - m_1x = 0$ & $y - m_2x = 0$, respectively.

Therefore, we can write

$$y^2 + \frac{2h}{b}xy + \frac{a}{b}x^2 \equiv (y - m_1x)(y - m_2x)$$

$$\Rightarrow y^2 + \frac{2h}{b}xy + \frac{a}{b}x^2 \equiv y^2 - (m_1 + m_2)x + m_1m_2x^2$$

Equating the coefficients of the like terms from both

sides we have

$$\left.\begin{aligned} m_1 + m_2 &= -\frac{2h}{b} \\ m_1 m_2 &= a/b \end{aligned}\right\} \quad \text{——— (3)}$$

From the figure, if $AB \perp AC$, i.e. if $\angle A = 90°$, then

$$a + b = 0 \quad \text{——— (A)}$$

If $AB \perp BC$, i.e., if $\angle B = 90°$, then

$$m_2\left(-\frac{l}{m}\right) = -1$$

$$\Rightarrow m_2 = \frac{m}{l} \quad \text{——— (4)}$$

Since, $y = m_2 x$ is one of the straight lines represented by (1), thus

$$ax^2 + 2hx(m_2 x) + b(m_2 x)^2 = 0$$

$$\Rightarrow ax^2 + 2hm_2 x^2 + bm_2^2 x^2 = 0$$

$$\Rightarrow a + 2hm_2 + bm_2^2 = 0 \quad [\because x^2 \neq 0]$$

$$\Rightarrow a + 2h\left(\frac{m}{l}\right) + b\left(\frac{m}{l}\right)^2 = 0 \quad [\text{using (4)}]$$

$$\Rightarrow a + \frac{2hm}{l} + \frac{bm^2}{l^2} = 0$$

$$\Rightarrow al^2 + 2hlm + bm^2 = 0 \quad \text{——— (B)}$$

Again if $BC \perp AC$, i.e. if $\angle C = 90^\circ$ then

$$m_1\left(-\frac{l}{m}\right) = -1$$

$$\Rightarrow \quad m_1 = \frac{m}{l}$$

Since, $y = m_1 x$ is one of the straight lines represented by ①, then

$$ax^2 + 2hx(m_1 x) + b(m_1 x)^2 = 0$$

$$\Rightarrow ax^2 + 2hm_1 x^2 + bm_1^2 x^2 = 0$$

$$\Rightarrow a + 2hm_1 + bm_1^2 = 0 \qquad [\because x^2 \neq 0]$$

$$\Rightarrow a + 2h\left(\frac{m}{l}\right) + b\left(\frac{m}{l}\right)^2 = 0$$

$$\Rightarrow al^2 + 2hml + bm^2 = 0 \qquad \text{——— ③}$$

Therefore, the required conditions that the triangle ABC will be right angled are given by

$$(a+b)(al^2 + 2hlm + bm^2) = 0 \qquad [\text{using } ④, ⑤, ③]$$

(Showed)

Problem:

If the equation $ax^2+2hxy+by^2+2gx+2fy+c=0$ represents a pair of straight lines equidistant from the origin, then prove that

(i) $f^4-g^4 = c(bf^2-ag^2)$

(ii) $h(g^2-f^2)=fg(a-b)$

Soln: i)

Given,

$$ax^2+2hxy+by^2+2gx+2fy+c=0 \quad \text{——— (1)}$$

Let the two straight lines represented by (1) are

$$l_1x+m_1y+n_1=0 \quad \text{——— (2)}$$

& $$l_2x+m_2y+n_2=0 \quad \text{——— (3)}$$

Therefore, we can write

$$ax^2+2hxy+by^2+2gx+2fy+c \equiv (l_1x+m_1y+n_1)(l_2x+m_2y+n_2)$$

$$\Rightarrow ax^2+2hxy+by^2+2gx+2fy+c \equiv l_1l_2x^2+l_2m_1xy+l_2n_1x + l_1m_2xy+m_1m_2y^2+m_2n_1y + l_1n_2x+m_1n_2y+n_1n_2$$

Equating the coefficients of the like terms from both sides we have,

$l_1 l_2 = a$; $l_1 m_2 + l_2 m_1 = 2h$; $m_1 m_2 = b$;

$l_1 n_2 + l_2 n_1 = 2g$; $m_1 n_2 + m_2 n_1 = 2f$; $n_1 n_2 = c$;

Given that lines ② & ③ are equidistant from the origin, then we have

$$\left| \frac{n_1}{\sqrt{l_1^2 + m_1^2}} \right| = \left| \frac{n_2}{\sqrt{l_2^2 + m_2^2}} \right|$$

$$\Rightarrow \frac{n_1^2}{l_1^2 + m_1^2} = \frac{n_2^2}{l_2^2 + m_2^2}$$

$$\Rightarrow n_1^2 (l_2^2 + m_2^2) = n_2^2 (l_1^2 + m_1^2)$$

$$\Rightarrow n_1^2 l_2^2 + n_1^2 m_2^2 = n_2^2 l_1^2 + n_2^2 m_1^2$$

$$\Rightarrow n_1^2 l_2^2 - n_2^2 l_1^2 = n_2^2 m_1^2 - n_1^2 m_2^2$$

$$\Rightarrow (n_1 l_2 + n_2 l_1)(n_1 l_2 - n_2 l_1) = (n_2 m_1 + n_1 m_2)(n_2 m_1 - n_1 m_2)$$

$$\Rightarrow (l_1 n_2 + l_2 n_1)(l_2 n_1 - l_1 n_2) = (m_1 n_2 + m_2 n_1)(n_2 m_1 - n_1 m_2)$$

$$\Rightarrow 2g\sqrt{(l_2 n_1 + l_1 n_2)^2 - 4 l_1 l_2 n_1 n_2} = 2f\sqrt{(m_1 n_2 + m_2 n_1)^2 - 4 m_1 m_2 n_1 n_2}$$

$$\Rightarrow 2g\sqrt{(2g)^2 - 4ac} = 2f\sqrt{(2f)^2 - 4bc}$$

$$\Rightarrow g\sqrt{4g^2 - 4ac} = f\sqrt{4f^2 - 4bc}$$

$\Rightarrow g\sqrt{g^2-ac} = f\sqrt{f^2-bc}$

$\Rightarrow g^2(g^2-ac) = f^2(f^2-bc)$

$\Rightarrow g^4 - g^2ac = f^4 - f^2bc$

$\Rightarrow f^4 - g^4 = f^2bc - g^2ac$

$\Rightarrow f^4 - g^4 = c(bf^2 - ag^2)$ ——— (4)

(Proved)

ii) Since equⁿ (1) represents a pair of straight line, so $\Delta = 0$

$\Rightarrow abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$\Rightarrow (ab-h^2)c = af^2 + bg^2 - 2fgh$

$\Rightarrow c = \dfrac{af^2+bg^2-2fgh}{ab-h^2}$

$\Rightarrow \dfrac{f^4-g^4}{bf^2-ag^2} = \dfrac{af^2+bg^2-2fgh}{ab-h^2}$ [using (4)]

$\therefore$

$\Rightarrow (f^4-g^4)(ab-h^2) = (bf^2-ag^2)(af^2+bg^2-2fgh)$

$\Rightarrow abf^4 - abg^4 - h^2f^4 + g^4h^2 = abf^4 - a^2f^2g^2 + b^2f^2g^2 - abg^4 - 2bf^3gh + 2afg^3h$

$$\Rightarrow a^2f^2g^2 - 2afg^3h + g^4h^2 = b^2f^2g^2 - 2bfg^3h + h^2f^4$$

$$\Rightarrow (afg)^2 - 2fg\cdot(hg^2) + (hg^2)^2 = (bfg)^2 - 2bfg(hf^2) + (hf^2)^2$$

$$\Rightarrow (afg - hg^2)^2 = (bfg - hf^2)^2$$

$$\Rightarrow afg - hg^2 = bfg - hf^2$$

$$\Rightarrow afg - bfg = hg^2 - hf^2$$

$$\Rightarrow h(g^2 - f^2) = fg(a-b)$$

(Proved)

Problem:

Show that one of the two lines represented by $ax^2 + 2hxy + by^2 = 0$ will coincide with one of the two lines given by $a_1x^2 + 2h_1xy + b_1y^2 = 0$ if

$$(ab_1 - a_1b)^2 = 4(a_1h - ah_1)(bh_1 - b_1h)$$

Soln:

Given, $ax^2 + 2hxy + by^2 = 0$ ——— (1)

& $a_1x^2 + 2h_1xy + b_1y^2 = 0$ ——— (2)

Let the common straight line of ① & ② be

$$y = mx$$

From ① we can write

$$ax^2 + 2hx^2m + bx^2m^2 = 0$$

$$\Rightarrow bm^2 + 2hm + a = 0 \quad [\because x^2 \neq 0] \quad \text{——— ③}$$

Similarly, from ② we can write

$$b_1m^2 + 2h_1m + a_1 = 0 \quad \text{——— ④}$$

Solving ③ & ④ we have

$$\frac{m^2}{2ha_1 - 2h_1a} = \frac{m}{ab_1 - a_1b} = \frac{1}{2h_1b - 2hb_1}$$

$$m = \frac{2(a_1h - ah_1)}{ab_1 - a_1b} \quad \& \quad m = \frac{ab_1 - a_1b}{2(h_1b - b_1h)}$$

Problem:

Prove that two of the straight lines represented by the equⁿ $ax^3 + bx^2y + cxy^2 + dy^3 = 0$, will be at right angles if $a^2 + ac + bd + d^2 = 0$

Solⁿ:

Given,

$$ax^3 + bx^2y + cxy^2 + dy^3 = 0 \quad \text{——— ①}$$

Here, equⁿ ① is a homogeneous equⁿ of degree-3. So, the given equⁿ will represent three straight lines that pass through the origin.

Let the equⁿ of the pair of straight lines perpendicular to each other be

$$x^2 + kxy - y^2 = 0 \quad \text{——— (2)}$$

& the equⁿ of the straight line be

$$ax - dy = 0 \quad \text{——— (3)}$$

Thus,

$$ax^3 + bx^2y + cxy^2 + dy^3 \equiv (x^2 + kxy - y^2)(ax - dy)$$

$$= ax^3 + akx^2y - axy^2 - dx^2y - dkxy^2 + dy^3$$

$$\Rightarrow ax^3 + bx^2y + cxy^2 + dy^3 = ax^3 + x^2y(ak - d) - xy^2(a + dk) + dy^3$$

Equating the coefficients of x^2y & xy^2 from both sides we get

$$b = ak - d$$

$$\Rightarrow k = \frac{b+d}{a}$$

& $c = -(a + dk)$

$$\Rightarrow c = -a - d\left(\frac{b+d}{a}\right)$$

$$\Rightarrow c = -a - \frac{bd + d^2}{a}$$

$$\Rightarrow c + a + \frac{bd + d^2}{a} = 0$$

$$\Rightarrow ac + a^2 + bd + d^2 = 0$$

$$\Rightarrow a^2 + ac + bd + d^2 = 0$$

(Proved)

Example:

If $ax^2+2hxy+by^2+2gx+2fy+c=0$ represents a pair of straight lines, then show that the area of the triangle formed by their bisectors and x-axis is

$$\frac{\sqrt{\{(a-b)^2+4h^2\}}}{2h}\cdot\frac{(a-g^2)}{(ab-h^2)}$$

Soln:

Given, $ax^2+2hxy+by^2+2gx+2fy+c=0$ ——— (1)

represents a pair of straight lines if

$$\Delta=\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}=0$$

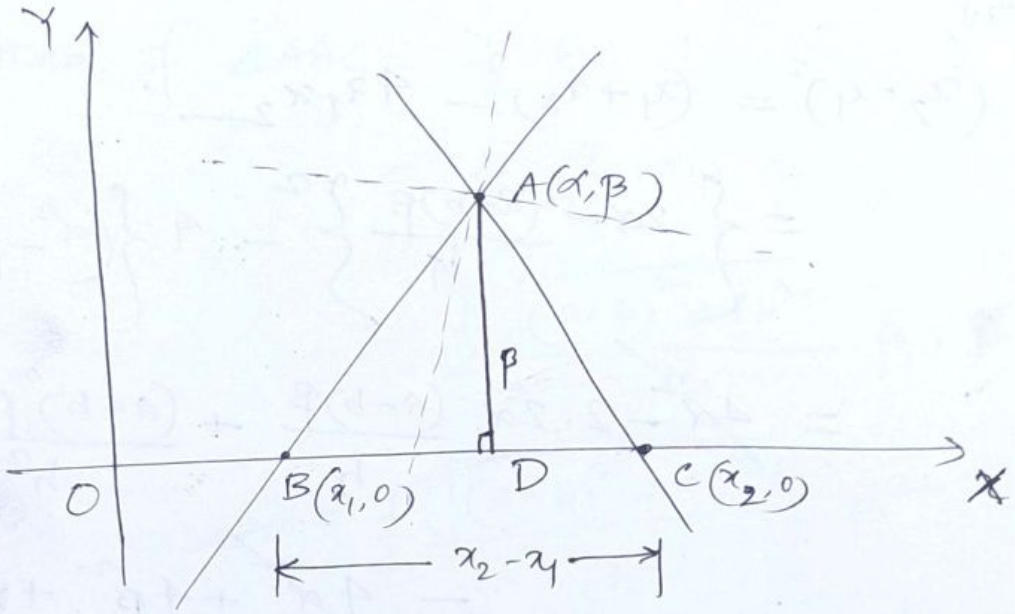
$$\Rightarrow abc+2fgh-af^2-bg^2-ch^2=0 \quad \text{——— (2)}$$

Let (α,β) be the point of intersection of the straight lines represented by (1), then

$$\alpha=\frac{hf-bg}{ab-h^2} \quad \& \quad \beta=\frac{hg-af}{ab-h^2}$$

We know, that the equn of bisectors

$$\frac{(x-\alpha)^2-(y-\beta)^2}{(x-\alpha)(y-\beta)}=\frac{a-b}{h} \quad \text{——— (3)}$$



The equⁿ of X-axis is $y = 0$, then from ③ we get

$$\frac{(x-\alpha)^2 - \beta^2}{(x-\alpha)(-\beta)} = \frac{a-b}{h}$$

$$\Rightarrow h\{x^2 - 2x\alpha + \alpha^2 - \beta^2\} = -\beta(x-\alpha)(a-b)$$

$$\Rightarrow hx^2 - 2x\alpha h + h(\alpha^2-\beta^2) + \beta(a-b)x - \alpha\beta(a-b) = 0$$

$$\Rightarrow hx^2 - \{2h\alpha - (a-b)\beta\}x + \{h(\alpha^2-\beta^2) - \alpha\beta(a-b)\} = 0 \quad \text{---- ④}$$

which is a quadratic equⁿ of x

Let the roots of equⁿ ④ be x_1 & x_2

$$\therefore x_1 + x_2 = \frac{2h\alpha - (a-b)\beta}{h} = 2\alpha - \frac{(a-b)\beta}{h}$$

$$\& \quad x_1 x_2 = \frac{h(\alpha^2-\beta^2) - \alpha\beta(a-b)}{h} = \alpha^2 - \beta^2 - \frac{(a-b)}{h}\alpha\beta$$

Now,

$$(x_2-x_1)^2 = (x_1+x_2)^2 - 4x_1x_2$$

$$= \left\{2\alpha - \frac{(a-b)\beta}{h}\right\}^2 - 4\left\{(\alpha^2-\beta^2) + \left(\frac{a-b}{h}\right)\alpha\beta\right\}$$

$$= 4\alpha^2 - 2\cdot 2\alpha\cdot\frac{(a-b)\beta}{h} + \frac{(a-b)^2\beta^2}{h^2}$$

$$- 4\alpha^2 + 4\beta^2 + \frac{4(a-b)}{h}\alpha\beta$$

$$= \frac{(a-b)^2}{h^2}\beta^2 + 4\beta^2 - \frac{4(a-b)\alpha\beta}{h} + \frac{4(a-b)\alpha\beta}{h}$$

$$= \frac{(a-b)^2\beta^2 + 4h^2\beta^2}{h^2}$$

$$= \frac{\{(a-b)^2 + 4h^2\}\beta^2}{h^2}$$

$$\therefore x_2 - x_1 = \frac{\sqrt{(a-b)^2 + 4h^2}}{h}\cdot\beta$$

From the figure, we can say that the vertices of the triangle ABC be $A(\alpha, \beta)$, $B(x_1, 0)$ & $C(x_2, 0)$ Also, the height of the ΔABC on BC is β.

$\therefore$ Area of $\Delta ABC = \frac{1}{2} \cdot BC \cdot AD$

$$= \frac{1}{2} \cdot (x_2 - x_1) \cdot \beta$$

$$= \frac{1}{2} \cdot \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \cdot \beta \cdot \beta$$

$$= \frac{1}{2} \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \cdot \beta^2$$

$$= \frac{1}{2} \cdot \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \cdot \left(\frac{hg - af}{ab - h^2} \right)^2$$

$$= \frac{1}{2} \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \left\{ \frac{h^2g^2 - 2hgaf + a^2f^2}{(ab-h^2)^2} \right\}$$

$$= \frac{1}{2} \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \left\{ \frac{h^2g^2 + a(af^2 - 2fgh)}{(ab-h^2)^2} \right\}$$

$$= \frac{1}{2} \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \left\{ \frac{h^2g^2 + a(abc - bg^2 - ch^2)}{(ab-h^2)^2} \right\}$$

[Using (2)]

$$= \frac{1}{2} \frac{\sqrt{(a-b)^2 + 4h^2}}{h} \left\{ \frac{h^2g^2 + a^2bc - abg^2 - ach^2}{(ab-h^2)^2} \right\}$$

$$= \frac{1}{2}\frac{\sqrt{(a-b)^2+4h^2}}{h}\left\{\frac{ac(ab-h^2)-g^2(ab-h^2)}{(ab-h^2)^2}\right\}$$

$$= \frac{1}{2}\frac{\sqrt{(a-b)^2+4h^2}}{h}\left\{\frac{(ab-h^2)(ac-g^2)}{(ab-h^2)^2}\right\}$$

$$= \frac{1}{2}\frac{\sqrt{(a-b)^2+4h^2}}{h}\cdot\frac{(ca-g^2)}{(ab-h^2)}$$

(Showed)

Example:

Prove that $x^2+6xy+9y^2+4x+12y-5=0$ represents a pair of parallel straight lines and the distance between them is $\frac{3\sqrt{10}}{5}$.

Sol^n:

Given, $$x^2+6xy+9y^2+4x+12y-5=0 \quad \text{——} (1)$$

$$\Rightarrow x^2+2(3y+2)x+(9y^2+12y-5)=0$$

$$\therefore x = \frac{-2(3y+2)\pm\sqrt{4(3y+2)^2-4.1(9y^2+12y-5)}}{2.1}$$

$$\Rightarrow x = -(3y+2) \pm \sqrt{(3y+2)^2 - (9y^2+12y-5)}$$

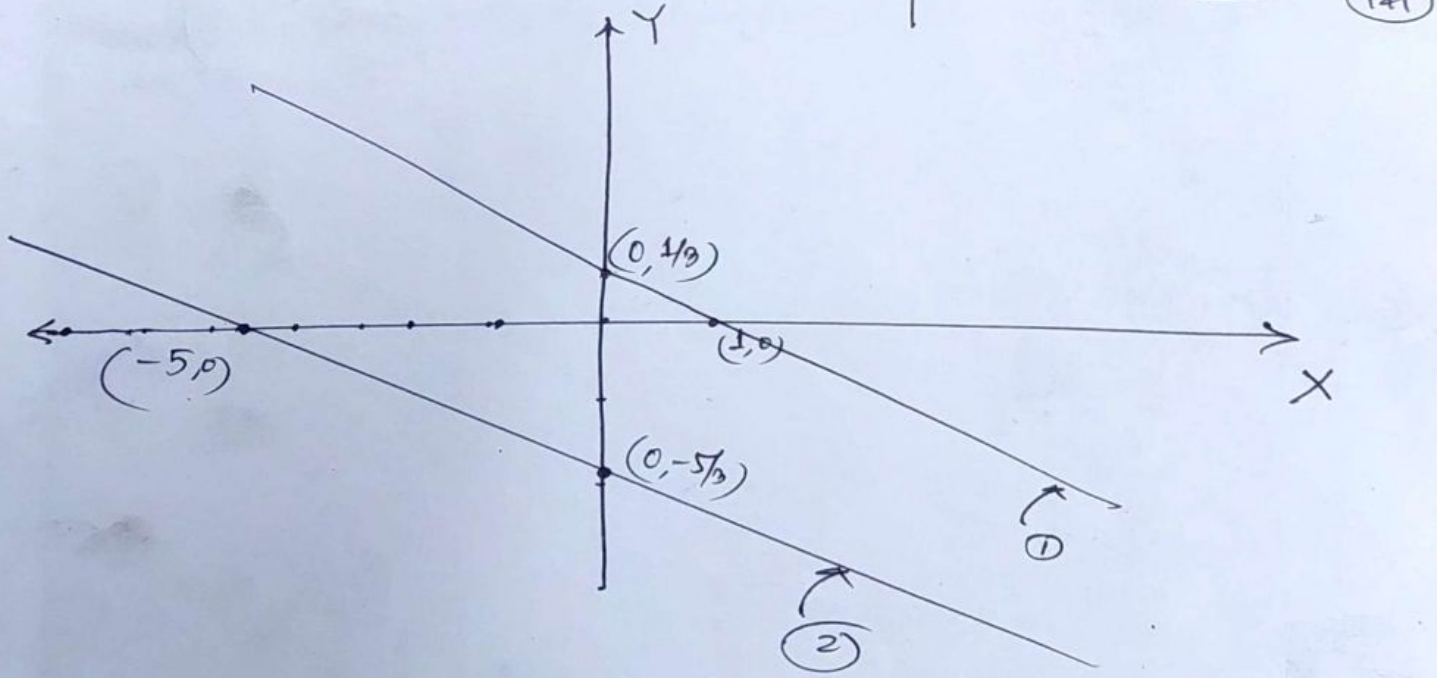
$$\Rightarrow x = -(3y+2) \pm \sqrt{9y^2+12y+4-9y^2-12y+5}$$

$$\Rightarrow x = -(3y+2) \pm 3$$

$\therefore x = -3y-2+3$ ——— (II) OR, $x = -3y-2-3$

$\Rightarrow x = -3y+1$ | $\Rightarrow x = -3y-5$

$\Rightarrow x+3y-1=0$ ——— (II) | $\Rightarrow x+3y+5=0$ ——— (II')



Slope of eqn (II), $m_1 = -1/3$

Slope of eqn (II'), $m_2 = -1/3$

So, the given eqn (I) represents a pair of parallel straight lines.

The distance between the two lines (11) & (14) is

$$d = \left| \frac{5-(-1)}{\sqrt{1^2+3^2}} \right| = \frac{6}{\sqrt{10}} = \frac{6\sqrt{10}}{10}$$

$$\Rightarrow d = \frac{3\sqrt{10}}{5}$$

Ans